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Lead Partner	OpenAIRE		
Authors	W. Hugo (KNAW-DANS) M. Katsamakis (OpenAIRE) P. Tsiavos (OpenAIRE)		
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Index of Terms

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Classes

Access Rule(s)	AssertionCollection	AssertionPolicy	ContextAssertion
Evaluation Detail	Evaluation Mapping	Licence	Named Constraint
OpenREL Evaluation Vocabulary	Parameter	Parameter Resolution Service	Policy Collection
Post-Action Duty	Pre-Action Duty	Process Rule(s)	RequestPolicy
ResolutionResult	ResolvedAssertion		

Properties

Access Type	Action Category	Applicable Endpoint	Applicable Scenario
Asserted By	Compared Individual	Compared Subject	Constraint Type
Created	Default Duty	Derivation Type	Duty
Encoded Rules	Evaluation Class	Evaluation Outcome	Expected Datatype
Has Evaluation	Has Evaluation Detail	Has Legal Basis	Has Policy
Interface Specification	Jurisdiction	Legal Code	Liability Limitation
Multiple Values	Named Constraint Reference	Object Rule Role	Obtain Consent
Operand Parameter	Parameter Name	Parameter Resolution Service	Policy
Policy Evaluation Outcome	Policy Status	Policy Text (Html)	Policy Text Resource
Rationale	Resolution Method	Resulting Policy	Source Relation
Subject Rights	Subject Rule Role	Support Count	Use
Validity Type	Warranty Notice	Was Derived From	

Concept Schemes

Evaluation Outcome Types	IDS Connector Security Profiles	Notification and Recording Types	OpenREL Policies
Parameter Binding Sources	Requestor Context Types	revenue-source-type-scheme	Scenario Types
Types of Access	Types of Agents	Types of Assertion	Types of Consent and Authorisation
Types of Named Constraints	Types of Policy Status	Types of Rule Set Validity	Types of Subject Rights
Types of Technical and Organisational Measures	Types of Usage	Types of Usage Purpose	

Individuals in Registries

Academic Use Only	Access Limitations	Access Token	Access Token Verification Endpoint
ActionsRegistry	Adaptation Status	Adaptation Status	Agent Affiliation Constraint
Agent Role	Allow Distribution Fee	Allow EU-wide Use	Asserted Researcher
Asset Licence List	Asset Publication Date	Assignee IRI	Assigner IRI
Authorised Endpoint IRIs	Authorised IP Addresses	Calculated Start Date	Calculated Start Date Constraint
Clearinghouse API	Consent is Required Prior to Access	Consent or Authorisation Required	Consent or Authorisation Verification
Consent Required on Legal Basis	Context Limitation	Data Specification IRI	Data Subject IRI
Delegation Token	Disallow Special Data Category	Distribute Unmodified	eIDAS Researcher Status
eIDAS Researcher Status	eIDAS Verified Identity	eIDAS Verified Researcher	Embargo Duration
End Date	End Date Constraint	Ethics Approval Required	External Institutional Recipient
Harvested Volume	Informed Consent Evidence IRI	Informed Consent Expiry Date	Infrastructure IRI
Institution IRI	Institutional Affiliation Required	Institutional Infrastructure Only	Intended Purpose IRI
IP Range	LeftOperandRegistry	Legal Basis	Location Limitation
Logging Specification	Managed Access Policies Only	Maximum Download Volume	Maximum Usage Count
NamedConstraintRegistry	Notification by Citation	Notification Duty	Notification in Asset
Notify Usage to a Clearinghouse	Obtain Ethics Approval	Open Access Policies Only	ParameterRegistry
Physical or Virtual Asset Location	Processing Infrastructure	Publisher IRI	Purpose Verification IRI
Record Usage in a Log	Revenue Permitted	Security Profile	Start Date
Start Date Constraint	Target IRI	Technical Usage Limitations	TOM Limitation
Unit of Measure	Usage Count Limit	Usage Volume Limit	Verified AI Agent
Verified Trust+			

Annex A: Core OpenREL Specification

OpenREL Core Specification and ODRL Profile - draft specification version 0.5
6 November 2026.

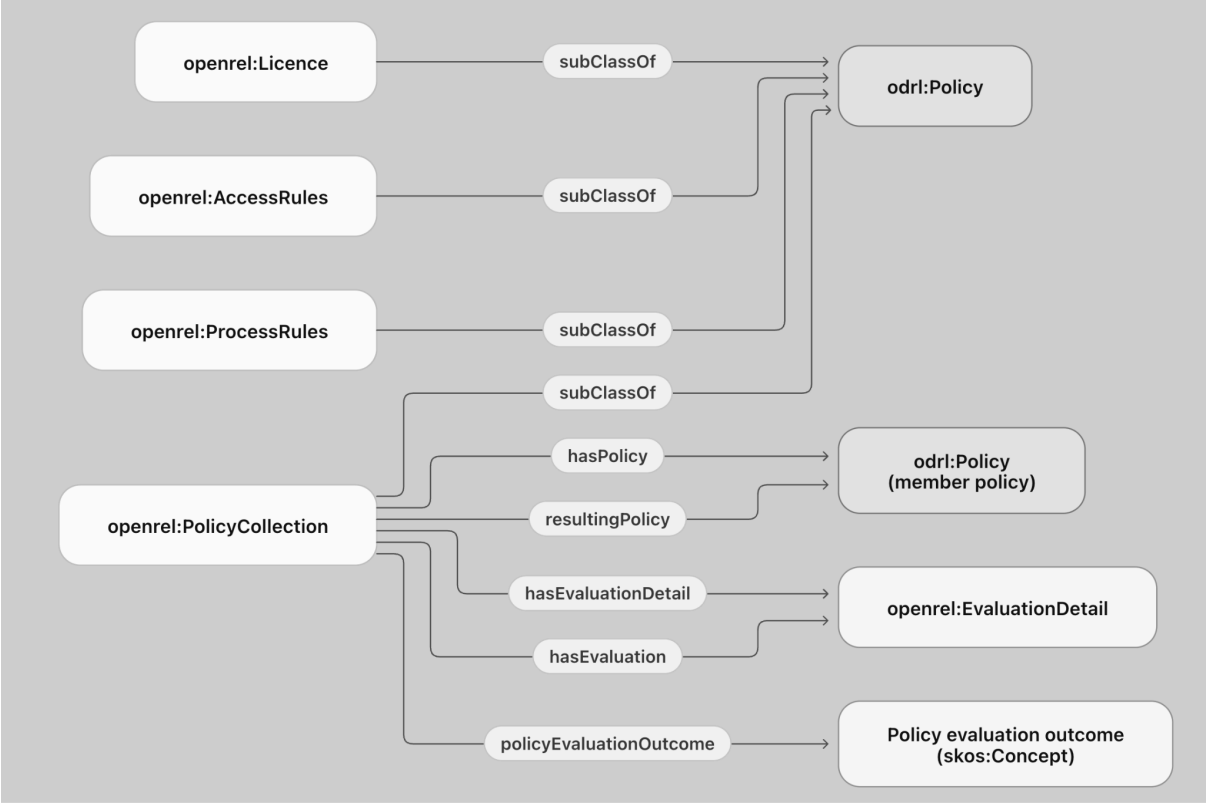
A.1.Policies

A.1.1 Introduction

ODRL policies express permissions, prohibitions and duties over assets and parties. OpenREL supplements the ODRL Policy model with policy categories distinguishing reusable licences, access requirements, process requirements and collections of referenced policies. The four new classes are subclasses of odrl:Policy, not replacements for the ODRL Policy types.

Figure A.1.1 presents the information model for OpenREL Policy Classes.

Figure A.1.1: OpenREL Policy and Evaluation Classes



A.1.2 Policy Classes

The ODRL policy classes include odrl:Set, odrl:Offer, odrl:Agreement, odrl:Request, odrl:Ticket, odrl:Privacy and odrl:Assertion. Their normative semantics remain those of the ODRL Information Model. OpenREL extensions are intended to distinguish policy content, metadata, evaluation and comparison outcomes, and provenance rather than the contractual lifecycle alone.

Table A.1.2: OpenREL Policy Classes/ Concept Scheme

Class	Purpose	Typical application/ Example

openrel:PolicyScheme (Concept Scheme)	OpenREL additionally defines the subclasses of Licence, Access Rules, and Process Rules	OpenREL policy classes are also SKOS Concepts.
openrel:Licence	Reusable rights and obligations of a licence	CC BY 4.0 applicable to published datasets
openrel:AccessRules	Conditions governing access outside licence terms	Repository terms, data-subject consent
openrel:ProcessRules	Procedural and platform access requirements	Registration, API rate limits
openrel:PolicyCollection	Reference multiple policies through a single policy resource	Repository-wide composite policy or negotiation record

A.1.2.1 Licence

IRI: openrel:Licence | Superclass: odrl:Policy

A reusable, generally non-negotiable Policy representing licence terms. A Licence MAY reference a legal-code representation. The corresponding property is specified in Section 3.

Example encoding (RDF TTL)

```
openrel:licenceExample a openrel:Licence ;
   dct:title "Example data licence"@en ;
   openrel:legalcode <https://example.org/licences/data/legalcode> .
```

A.1.2.2 AccessRules

IRI: openrel:AccessRules | Superclass: odrl:Policy

Access conditions associated with a repository, institution, service or data subject, separate from the licence governing use. The example encoding references a constraint that requires requestors to be an eIDAS-verified researcher.

```
openrel:repositoryAccess a openrel:AccessRules ;
   dct:title "Repository access conditions"@en ;
      odrl:permission [
      odrl:action openrel:read ;
       odrl:constraint <11-eidas-researcher>
   ]
   .
```

A.1.2.3 ProcessRules

IRI: openrel:ProcessRules | Superclass: odrl:Policy

Formalities imposed by a software platform or dissemination channel. A process policy can be expressed through ODRL or OpenREL rules and constraints; this example shows only the structural pattern. The encoding example includes a named constraint (defined later) to enforce Trust Plus certification for access by an automated agent, and the requirement to log instances of use at a clearinghouse endpoint.

```
openrel:platformProcess a openrel:ProcessRules ;
   dct:title "Platform download procedures"@en ;
   odrl:permission [
       odrl:action openrel:use ;
       odrl:constraint openrel:trustPlus ;
       odrl:duty [
           odrl:action openrel:notify ;
           odrl:constraint openrel:notifyClearingHouse
       ]
   ] .
```

A.1.2.4 PolicyCollection

IRI: openrel:PolicyCollection | Superclass: odrl:Policy

A Policy Collection references constituent Policy IRIs using openrel:hasPolicy, with domain openrel:PolicyCollection and range odrl:Policy. The encoding example indicates a typical combination of three policies: a licence, governing the rights of the owner of the asset, a policy that determines the basis of access to the repository where the asset is hosted, and a policy that manages the rules of the dissemination platform in use by the repository.

```
openrel:combinedPolicy a openrel:PolicyCollection ;
   dct:title "Dataset policy bundle"@en ;
   openrel:hasPolicy openrel:licenceExample,
       openrel:repositoryAccess, openrel:platformProcess .
```

A PolicyCollection has a constraint in respect of cardinality: the openrel:hasPolicy predicate MUST indicate at least two policies. This cardinality constraint is defined in openrel:PolicyCollectionShape, a SHACL constraint.

A.1.3 Policy Properties

The following table specifies supplementary OpenREL policy metadata properties and the constituent-policy relationship. The metadata properties share an OWL union domain comprising the four OpenREL Policy subclasses; openrel:hasPolicy has the narrower domain openrel:PolicyCollection. Many of the properties mirror those defined by SPDX[1] for licence metadata, or by Creative Commons[2].

Table A.5.6.12: OpenREL Policy-Related Properties

Property	Domain	Range	Group / notes
openrel:accessType	Policy subclasses[1]	skos:Concept openrel:typeAccess	Access classification applicable to the policy.
openrel:policyStatus	Policy subclasses[1]	skos:Concept openrel:typeStatus	Coverage/status.
openrel:jurisdiction	Policy subclasses[1]	dct:Location	Coverage; subproperty of cc:jurisdiction.
openrel:validityType	Policy subclasses[1]	skos:Concept openrel:typeValidity	Coverage/temporal classification.
openrel:liabilityLimitation	Policy subclasses[1]	rdf:langString	Disclaimers.
openrel:warrantyNotice	Policy subclasses[1]	rdf:langString	Warranty statement.
openrel:legalcode	Policy subclasses[1]	rdfs:Resource	Alternative representations - legal code
openrel:policyTextHtml	Policy subclasses[1]	rdfs:Resource	Alternative representations - human-readable HTML
openrel:policyText	Policy subclasses[1]	rdfs:Resource	Alternative representations - human-readable text
openrel:encodedRules	Policy subclasses[1]	odrl:Policy	Machine-readable representations, using e.g. OpenREL.
openrel:hasPolicy	openrel:PolicyCollection	odrl:Policy	Policy Collection composition.
openrel:resultingPolicy	openrel:PolicyCollection	odrl:Policy	References a policy derived by combining the applicable requirements of the policies in a PolicyCollection.

[1] SPDX - https://github.com/spdx/license-list-XML/blob/main/DOCS/license-fields.md
[2] Creative Commons - https://creativecommons.org/

openrel:subjectRights	Policy subclasses[1]	skos:Concept openrel:typeSubjectRights	Identifies a right available to a subject in relation to an Asset or processing activity governed by an OpenREL Policy

[1] Policy subclasses = OWL union of openrel:Licence, openrel:AccessRules, openrel:ProcessRules and openrel:PolicyCollection.

A.1.4 Policy Collection Composition

The constituent-policy relationship is expressed by openrel:hasPolicy, an owl:ObjectProperty and skos:Concept. Its domain is openrel:PolicyCollection and its range is odrl:Policy.

```
openrel:hasPolicy a owl:ObjectProperty, skos:Concept ;
   rdfs:domain openrel:PolicyCollection ;
   rdfs:range odrl:Policy .
```

A.1.5 Context and Assertions

OpenREL defines a number of contexts that collectively determine whether a policy of any type can be satisfied, for example, by a request for access. In principle, an Assertion can provide contextual claims for a Request. A separate specification extends the Assertion and Request policy subclasses defined in ODRL to allow resolution of contextual parameters for purposes of evaluation.

A.1.6 Conformance and Profile Considerations

OpenREL Policy instances retain the semantics and constraints of their ODRL superclass. The ODRL Profile extension rules permit additional Policy subclasses subject to their disjointness requirements. Supplementary metadata properties do not constitute a separately sanctioned ODRL Profile extension mechanism.

A.1.7 Prefixes Used in Examples

```
@prefix openrel: <http://www.w3id.org/ns/openrel/0/> .
@prefix odrl: <http://www.w3.org/ns/odrl/2/> .
@prefix dct: <http://purl.org/dc/terms/> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix cc: <http://creativecommons.org/ns#> .
```

A.1.8 Applicable Concept Schemes

The following concept schemes, defined by OpenREL, are applicable to Policy-related predicates and properties.

A.1.8.1 Access Type

Concept scheme IRI: openrel:typeAccess

The Access Types concept scheme defines categories of access applicable to resources. These concepts can be referenced using openrel:accessType in OpenREL policies.

Table A.1.8.1: openrel:typeAccess Concept Scheme

Concept IRI	Preferred Label	Definition	External Mappings

openrel:open-access	Open Access	A resource that is immediately and permanently available online, free of financial and technical barriers.	skos:exactMatch r3d:open, coar:c_abf2
openrel:embargoed	Embargoed	A resource for which access is restricted until a specified release date, after which it becomes openly accessible.	skos:exactMatch r3d:embargoed, coar:c_f1cf
openrel:restricted	Restricted	A resource available subject to access restrictions, such as authentication, authorisation, approval or community membership.	skos:exactMatch r3d:restricted, coar:c_16ec
openrel:closed	Closed	A resource whose metadata is available but whose content is not accessible.	skos:exactMatch r3d:closed, coar:c_14cb
openrel:managed	Managed	A resource accessible to users who satisfy predefined rules designed to protect the interests of the subjects. Eligibility can be determined before an access request is submitted.	No external mapping defined

Specification note: openrel:embargoed is retained for interoperability with COAR and re3data. In the OpenREL model, an embargo is primarily a temporal constraint that may apply to other access types, rather than an independent access category.

A.1.8.2 Policy Status Types

Concept scheme IRI: openrel:typeStatus

The Policy Status Types concept scheme identifies the lifecycle status of an OpenREL Policy and, for active policies, its intended scope of applicability.

Table A.1.8.1: openrel:typeStatus Concept Scheme

Concept IRI	Preferred label	Definition	Broader concept
openrel:typeDraft	Draft Policy	A policy that is still in draft status. Private by default.	-
openrel:typeReview	Policy in Review	A policy that is being reviewed. Private by default.	-
openrel:typeTemplate	Policy Template	A policy that is used as a template, with placeholders for parameters.	-
openrel:typeActive	Active Policy	A policy that is active. It can be private, custom, or global.	-
openrel:typePrivateActive	Active Policy - Private	A policy that is active but applicable only in a specific private context. For example, the assigner, assignee, and asset have all been explicitly defined.	openrel:typeActive
openrel:typeCustomActive	Active Policy - Custom	A policy that is active but applicable only in a specific custom context. For example, the assigner and asset collection may have been explicitly defined, typically for a repository or collection.	openrel:typeActive

openrel:typeGlobalActive	Active Policy - Global	A policy that is active and applicable globally. Typically, the assigner, assignee, and asset or asset collection are not defined.	openrel:typeActive

The openrel:policyStatus predicate allows multiple values, for example openrel:typeReview, openrel:typeTemplate CAN both be true for a policy.

A.1.8.3 Validity Type

Concept scheme IRI: openrel:typeValidity

The Validity Types concept scheme identifies qualifications affecting the applicability or scope of an OpenREL Policy or Licence.

Table A.1.8.3: openrel:typeValidity Concept Scheme

Concept IRI	Preferred label	Definition	External mapping
openrel:patentFree	Patent Free	The licence applies only to patent claims that a contributor can license and that are necessarily infringed by the contribution, alone or in combination with the licensed material.	skos:exactMatch dalicc:patentFree
openrel:irrevocable	Irrevocable	The licence cannot be terminated for any reason, or cannot be terminated for convenience but may still be terminated for breach.	skos:exactMatch dalicc:irrevocable
openrel:perpetual	Perpetual	A perpetual licence allows the licensed work to be used indefinitely.	skos:exactMatch dalicc:perpetual
openrel:worldwide	Worldwide	The licence terms are valid throughout the world.	skos:exactMatch dalicc:worldwide
openrel:community	Community	The policy or licence applies to a specific community.	-
openrel:custom	Custom	The policy or licence applies only to a specific institution, collection, catalogue, resource or asset.	-

Take note of the following:

- Patent Free: The DALICC definition describes which patent claims are covered by a licence, rather than necessarily asserting that an asset is free of patent rights. The preferred label may therefore be misleading.
- Irrevocable: The definition combines two different legal situations: unconditional irrevocability and irrevocability subject to termination for breach. These may eventually require separate concepts.
- Validity dimensions: The scheme combines temporal validity (perpetual), territorial scope (worldwide), community applicability (community), resource-specific applicability (custom) and patent-related conditions (patentFree). These are not mutually exclusive and may be used together.

A.1.9 References

W3C ODRL Information Model, Policy section (https://www.w3.org/TR/odrl-model/#policy); OpenREL Policies guidance; ODRL Profile extension rules.

A.2. Rules

OpenREL extends the existing ODRL Rules (odrl:Duty) with subclasses to account for temporal aspects of duties.

A.2.1 Pre-Action and Post-Action Duties

OpenREL defines two specialised subclasses of `odrl:Duty` to distinguish obligations according to their timing relative to the execution of an Action.

Table A.2.1: OpenREL Duty Subclasses

Class	Superclass	Description
openrel:PreDuty	odrl:Duty	A duty that MUST be fulfilled before the associated Action can be executed.
openrel:PostDuty	odrl:Duty	A duty that MUST be fulfilled after execution of the associated Action, subject to the conditions specified by the Policy.

A Duty MAY be classified as both `openrel:PreDuty` and `openrel:PostDuty` when its obligations apply at both stages. The two classes are therefore not mutually exclusive.

A PreDuty that remains unsatisfied MAY result in a `CONDITIONAL` evaluation outcome if fulfilment is still possible. An unsatisfiable mandatory condition results in `DENY`. A PostDuty does not ordinarily prevent initial execution, although subsequent non-compliance may constitute a policy violation. The relation between policy comparisons, the extent to which pre-duty and post-duty obligations have been satisfied, and compatibility of actions and constraints are discussed in detail in the section on ‘Comparison and Evaluation’.

OpenREL does not introduce additional properties for these subclasses. Existing ODRL properties remain applicable.

A.2.1.1 Pre Duty

IRI: openrel:PreDuty | Superclass: odrl:Duty

A Duty that MUST be fulfilled before the associated Action may be executed. A PreDuty represents a prerequisite for executing an Action.

A.2.1.1 Post Duty

IRI: openrel:PostDuty | Superclass: odrl:Duty

A Duty that MUST be fulfilled after execution of the associated Action, within the period or conditions specified by the applicable Policy. A PostDuty does not, by itself, prevent initial execution of the associated Action. Failure to fulfil it MAY constitute a policy violation or trigger consequences defined by the Policy.

A.3. Actions

A.3.1 Scope and Purpose

An Action denotes an operation that may be permitted, prohibited, or required as a duty under an ODRL Rule. OpenREL provides additional Action terms for research-data access, licensing, preservation, notifications, and related rights management. This section describes the Action

individuals, supporting vocabulary, and its relationship to other classes and properties. Actions can be subject to Constraints, these and their associated Operands are specified separately.

OpenREL favours general Actions refined by Constraints, Parameters, and supporting vocabulary over a proliferation of narrowly specialised Actions. For example, the recipient, purpose, target, or content of a notification may be specified through a single Action by varying the relevant parameters and/ or vocabulary..

A.3.2 ODRL Profile Alignment

An additional Action MUST be defined as an individual of odrl:Action, not as a subclass of odrl:Action. It MAY specify an odrl:includedIn parent Action and MAY use odrl:implies to express an action dependency, subject to the ODRL semantics. Every OpenREL Action introduced by the Profile is also declared a skos:Concept, with a human-readable label and definition.

The canonical OpenREL namespace is http://www.w3id.org/ns/openrel/0/. The prefix openrel: used throughout this section expands to this namespace. The ODRL namespace is http://www.w3.org/ns/odrl/2/. The encoding example below illustrates this approach.

```
@prefix openrel: <http://www.w3id.org/ns/openrel/0/> .
@prefix odrl: <http://www.w3.org/ns/odrl/2/> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .

openrel:archive a odrl:Action, skos:Concept ;
   skos:prefLabel "Archive"@en ;
   skos:definition "Store a resource in a long-term repository or archive for preservation
purposes."@en ;
   odrl:includedIn odrl:use .
```

A.3.3 Actions Registry

Table A.3.3 lists the 22 OpenREL Action individuals in the aligned Actions vocabulary, these individuals are also part of an OpenREL Actions Registry.

Registry IRI: openrel:ActionsRegistry

Table A.3.3 OpenREL Actions Registry

Action IRI	Label	Definition	Included in
openrel:anonymize	Anonymize	Transform data so that specific subject entities and their sensitive or privileged data cannot be identified.	odrl:use
openrel:applyLicence	Apply Licence	Attach a recognised licence to a resource, defining the terms for its use and distribution.	odrl:use
openrel:archive	Archive	Store a resource in a long-term repository or archive for preservation purposes.	odrl:use
openrel:assentTo	Assent To	An agent using the resource assents to/ acknowledges one or more duties.	-
openrel:attribute	Attribute	Credit the original creator or rights holder when using a resource.	odrl:use
openrel:commercialise	Commercialise	To use an Asset for commercial purposes or financial benefit.	-
openrel:costRecovery	Cost Recovery	To recover costs associated with permitted use or distribution of an Asset.	-
openrel:delete	Delete	Permanently remove a resource and/ or any copies of it.	odrl:use

openrel:derive	Derive	To create a new derivative asset from one or more assets and to edit or modify the derivative.	odrl:use
openrel:distribute	Distribute	Transfer, forward, supply, or share a resource with third parties. Encompasses sharing, disseminating, republishing, and redistributing.	odrl:use
openrel:encrypt	Encrypt	Transform data so that specific subject entities and their sensitive or privileged data cannot be identified.	odrl:use
openrel:log	Log	Record an event to a designated logging or audit system.	odrl:use
openrel:modify	Modify	Alter or adapt a resource resulting in a new (minor) version that is regarded as the same object, but updated or improved.	odrl:use
openrel:notify	Notify	Send a notification to a defined endpoint upon performing an action.	odrl:use
openrel:obtainEthicsApproval	Obtain Ethics Approval	To obtain documented approval or clearance from a competent research ethics committee or other authorised ethics review body for a specified activity involving an Asset.	odrl:use
openrel:promote	Promote	To use an Asset for promotional purposes.	-
openrel:read	Read	Read-only access to a resource without modification, redistribution, or derivation.	odrl:use
openrel:reproduce	Reproduce	Copy, download, or replicate a resource without modification, in any form.	odrl:use
openrel:sensitiveDataConsent	Sensitive Data Consent	Acquire and record informed consent from the data subject prior to, after, or both prior to and after processing, collection, or sharing of personal data, specifically in the case of sensitive data.	odrl:obtainConsent
openrel:shareAlike	Share Alike	The Asset or its derivatives may be distributed without modification of its original Policy.	-
openrel:trademarkUse	Trademark Use	The license grants/ does not grant permission to use the trade names, trademarks, service marks, or product names of the Licensor, except as required for reasonable and customary use in describing the origin of the Work and reproducing the content of the notice file.	-
openrel:use	Use	To use the Asset	-

A.3.4 Properties Associated with Actions

OpenREL declares supplementary properties applicable to Actions.

Table A.3.4: OpenREL Action Property Extensions

Property	Label	Definition in source
openrel:actionCategory	Action Category	Some Actions are typically applicable to specific Rules, for example attribution is typically a Duty (openrel:PostDuty).
openrel:defaultDuty	Default Duty	The Action typically implies this Duty.

A.3.5 Mappings and Semantic Assertions

OpenREL records comparisons between Actions as independent mapping individuals (‘Evaluation Records’). The compared Actions remain unmodified. An Evaluation Record identifies the compared subject and target, the comparison class (odrl:Action), the relationship or outcome, the assertion type, and the provenance of the assertion.

OpenREL defines a number of assertion types, and these are discussed in detail in the next section. In ODRL, Actions can define relationships to other Actions, and OpenREL accommodates this as described below.

A relationship explicitly defined by an authoritative source vocabulary, such as odrl:includedIn or odrl:implies, can be classified as a semantic assertion. This classification describes the provenance and derivation of the relationship. OpenREL allows such relationships to be defined explicitly, and to be combined with other types of relationship assertions. As discussed in the next section on comparisons and evaluation, the mappings created in this way (see example) can be added to a graph for reasoning and evaluation of compatibility between policies.

```
openrel:exampleActionComparison
   a openrel:EvaluationMapping ;
   openrel:comparedSubject odrl:print ;
   openrel:comparedIndividual odrl:reproduce ;
   openrel:evaluationClass odrl:Action ;
   openrel:evaluationOutcome odrl:includedIn
```

Relationships MUST be checked against the authoritative source vocabulary before being classified as semantic assertions. Read more about these comparisons and evaluation mappings in the section on Evaluation.

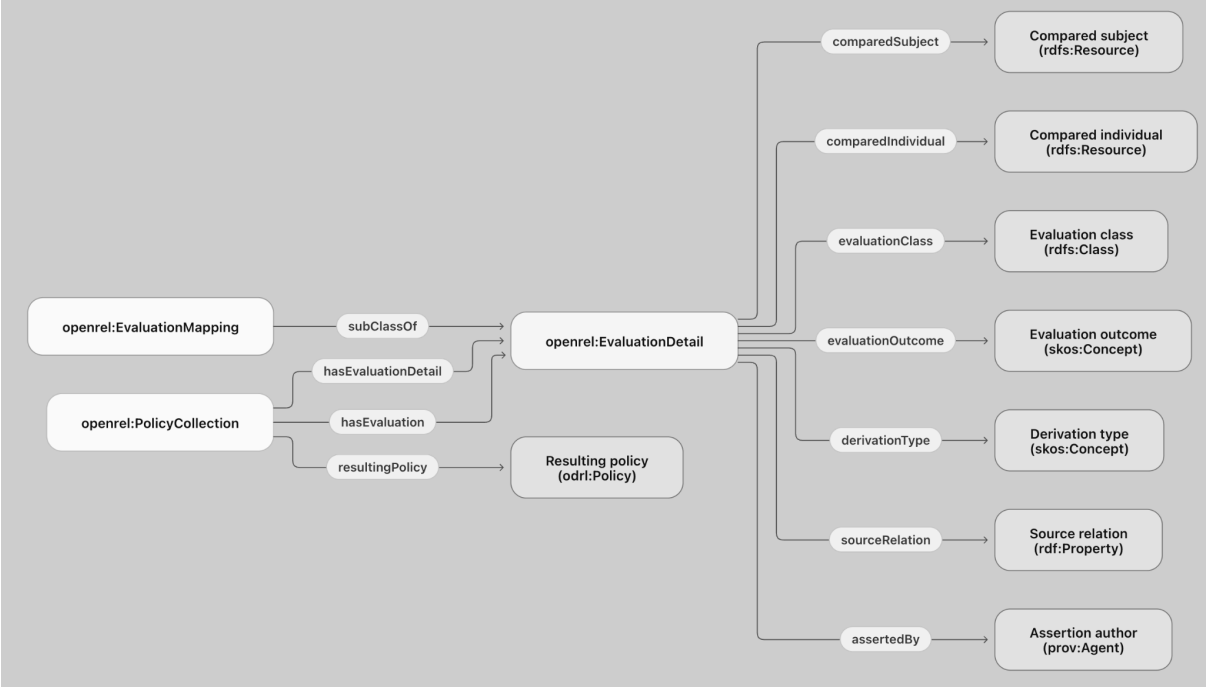
A.4. Comparison and Evaluation

OpenREL provides a vocabulary for recording directed graph comparisons between RDF resources, including ODRL Actions, Constraints and Policies. An evaluation record identifies the compared resources, the class of comparison, its asserted relationship or outcome, and the evidence or derivation supporting that assertion. Comparison records are maintained separately from the vocabulary definitions and do not modify the compared resources, and, as such, can be maintained as a knowledge graph that will become increasingly accurate in its comparisons.

This section specifies the RDF representation of evaluation records, and discusses how these are applied to comparisons between Actions and between Constraints. The calculation of request decisions, policy aggregation and enforcement is addressed separately in the Resolution and Evaluation Procedure section. In particular, an asserted Action relationship is not, for example, a direct PERMIT or DENY decision in respect of an access request for a specific Asset.

Figure A.4 presents Evaluation-related OpenREL classes and the relationships between them.

Figure A.4: Evaluation Information Model



A.4.1 Vocabulary Namespace

The OpenREL evaluation vocabulary is supplementary to ODRL. It does not redefine ODRL classes, properties or individuals, and does not assert that supplementary evaluation properties constitute ODRL Profile extensions.

```
@prefix openrel: <https://w3id.org/openrel/> .
@prefix odrl:    <http://www.w3.org/ns/odrl/2/> .
```

A.4.2. Evaluation Classes

Table A.4.2: OpenREL Evaluation Classes

Class	Superclass	Definition
openrel:EvaluationScheme (Concept Scheme)	-	Vocabulary for recording comparisons of policies, actions, constraints and other resources, including assertion provenance and outcomes. Classes below are also SKOS concepts.
openrel:EvaluationDetail	-	Record of a directed comparison between two resources, identifying the comparison class and asserted outcome.
openrel:EvaluationMapping	openrel:EvaluationDetail	Reusable mapping or relationship between two resources.

An EvaluationMapping is an EvaluationDetail object, and an EvaluationDetail object MAY contain multiple EvaluationMappings. Evaluations (EvaluationMappings) MAY concern Actions, Constraints, Policies or other resources. The direction of comparison is significant for asymmetric relationships, such as odrl:includedIn.

A.4.2.1 Evaluation Scheme

IRI: openrel:EvaluationScheme | Superclass: N/A

Vocabulary for recording comparisons of policies, actions, constraints and other resources, including assertion provenance and outcomes. Evaluation-related classes are also SKOS concepts.

A.4.2.2 Evaluation Detail

IRI: openrel:EvaluationDetail | Superclass: N/A

An Evaluation Detail provides the record of a directed comparison between two resources, identifying the comparison class and asserted outcome. Records of a pairwise comparison do not itself determine the aggregate PolicyCollection outcome.

A.4.2.3 Evaluation Mapping

IRI: openrel:EvaluationMapping | openrel:EvaluationDetail

A reusable Evaluation Detail representing a previously established relationship between two RDF resources. Mappings MAY be derived from e.g. semantic assertions, corpus observations, or human review. Their evidential strength depends on the derivation method and supporting provenance. See Table A.4.2.6.

A.4.2.4 PolicyCollection Evaluation

An `openrel:PolicyCollection` MAY contain one or more evaluation records describing comparisons between its constituent Policies. Such evaluations are represented by instances of `openrel:EvaluationDetail`, associated with the PolicyCollection through `openrel:hasEvaluationDetail`.

The overall outcome of a policy-level comparison is recorded using `openrel:policyEvaluationOutcome`. Its value MUST be one of `PERMIT`, `CONDITIONAL` or `DENY`, as defined by the OpenREL evaluation-outcome vocabulary. The mechanisms for comparing policies, and determining an outcome is discussed in the section on Resolution and Evaluation Procedure.

Table A.4.2.4: OpenREL PolicyCollection Properties related to Evaluation

Property	Domain	Range	Description
`openrel:hasEvaluationDetail`	`openrel:PolicyCollection`	`openrel:EvaluationDetail`	References an evaluation record associated with the PolicyCollection.
`openrel:policyEvaluationOutcome`	`openrel:PolicyCollection`	`skos:Concept Openrel: typeEvaluationOutcome`	Records the overall outcome of the policy-level comparison.

An Evaluation Detail MAY be supported by one or more `openrel:EvaluationMapping` instances that record pairwise comparisons between Actions, Constraints or other policy elements. The outcomes of these mappings provide evidence for the policy-level decision but do not independently determine it.

A PolicyCollection MAY have multiple evaluations, for example when different requests, contexts or combinations of Policies are assessed.

A.4.2.5 Evaluation-Related Properties

Table A.4.2.5: OpenREL Properties related to Evaluation

Property (openrel:)	Schema characteristic	Purpose

hasEvaluation	EvaluationDetail (range)	Links a resource to an evaluation involving it.
comparedSubject	EvaluationDetail (domain)	Source resource of the directed comparison.
comparedIndividual	EvaluationDetail (domain)	Target resource of the directed comparison.
evaluationClass	rdfs:Class (range)	RDF class of the compared resources, e.g. odrl:Action.
evaluationOutcome	EvaluationDetail (domain)	Asserted relationship or comparison outcome.
derivationType	skos:Concept (range) openrel:typeAssertion	Assertion method or provenance category.
subjectRole	rdfs:Class (range)	ODRL Rule class associated with the subject.
objectRole	rdfs:Class (range)	ODRL Rule class associated with the target.
sourceRelation	EvaluationDetail (domain)	Relationship as expressed in the original source.
rationale	xsd:string (range)	Human-readable explanation or evidence.
support	xsd:nonNegativeInteger (range)	Count of supporting source records.
assertedBy	prov:wasAttributedTo subproperty	Agent responsible for the assertion.
created	xsd:dateTime (range)	Creation timestamp; dct:created is also available.
wasDerivedFrom	EvaluationDetail (domain)	Source IRI or legacy textual description.

Properties are defined in the OpenREL schema with rdfs:label and skos:definition. The table abbreviates the RDF domain and range declarations; unspecified ranges are intentionally not constrained in the schema.

The property openrel:derivationType references an assertion-type concept from openrel:typeAssertion (see A.4.2.3). Relevant categories include semantic, deterministic, corpus-derived, probabilistic and human assertions, as defined by that scheme. The evaluation schema does not reproduce those concepts.

A semantic assertion reproduces an explicitly asserted relationship from an authoritative source vocabulary. For example, an evaluation may refer to odrl:includedIn as its evaluationOutcome while identifying the source vocabulary through provenance. Referencing an ODRL predicate as an outcome does not create an RDF triple between the two compared Actions.

```
openrel:exampleMapping a openrel:EvaluationMapping ;
   openrel:comparedSubject odrl:print ;
   openrel:comparedIndividual odrl:reproduce ;
   openrel:evaluationClass odrl:Action ;
   openrel:evaluationOutcome ordl:includedIn ;
   openrel:derivationType openrel:assertionSemantic .
```

These semantic assertions are validated in the source schemas, and as such, are considered to take precedence and effectively negate or nullify any other assertions of an evaluation outcome between two actions or constraints.

A.4.2.6 Assertion Types

Concept scheme IRI: `openrel:typeAssertion`

The Assertion Types concept scheme identifies how comparisons and mappings between resources have been established or derived.

Table A.4.2.6: openrel:typeAssertion Concept Scheme

Concept IRI	Preferred label	Definition
`openrel:assertionDeterministic`	Deterministic Assertion	An assertion produced by applying a defined, reproducible procedure that yields the same result for the same inputs.
`openrel:assertionCorpus`	Corpus-Derived Assertion	An assertion derived from analysing a corpus of policy or licence encodings and identifying relationships and common co-occurences within that corpus.
`openrel:assertionProbabilistic`	Probabilistic Assertion	An assertion inferred by a method that produces an estimated or uncertain relationship, for example through vector similarity or generative AI.
`openrel:assertionSemantic`	Semantic Assertion	A comparison derived directly from an explicitly defined semantic relationship between resources in an authoritative source vocabulary.
`openrel:assertionHuman`	Human Assertion	A comparison derived by a human through inspection of two policies, actions, or constraints.

A.4.2.7 Mapping Conflicts and Preferred Assertions

Each mapping MAY include an asserting agent, source relation, supporting observation count, rationale, and provenance. Multiple evaluation records MAY concern the same pair of resources and MAY contain contradictory outcomes from different sources. The schema does not resolve such conflicts automatically.

In practice, semantic assertions are considered authoritative, while human assertions are considered to be more reliable than others, should any duplicate assertions be available for the same pairwise comparison. The number of assertions from independent sources that agree on the evaluation outcome also strengthens the confidence in the assertion. Based on this:

- If a semantic assertion is present for a pairwise comparison, that assertion MUST be used even though other assertions for the same object-subject pair may be available.
- If a semantic assertion is not present for an object-subject pair, and a human assertion is available, that assertion SHOULD be used by preference.
- If multiple assertions are available for an object-subject pair, none of which are semantic or human in origin, the most common assertion MAY be considered to be applicable.
- A support count MUST NOT be interpreted, on its own, as proof of compatibility.

A.4.3. Recording a Mapping

The following example illustrates the structure of an existing corpus-derived Action comparison. The outcome records a source assertion; it is not a conclusion that the two rules are logically compatible (or not).

```
<https://w3id.org/openrel/reasoner/assertion-0004>
   a openrel:EvaluationMapping ;
   openrel:comparedSubject <http://creativecommons.org/ns#DerivativeWorks> ;
   openrel:comparedIndividual odrl:copy ;
   openrel:evaluationClass odrl:Action ;
   openrel:evaluationOutcome openrel:contradicts ;
   openrel:derivationType openrel:assertionCorpus ;
   openrel:subjectRole odrl:Prohibition ;
   openrel:objectRole odrl:Permission ;
   openrel:support 6 ;
   openrel:rationale "Co-occurrence in 6 of 336 licences." .
```

The example is illustrative and omits additional source metadata. The evidence supports observed co-occurrence object-subject pairs in a collection of policies, not necessarily permission, legal compatibility or logical entailment.

A.4.4. Provenance

Each mapping MAY include an asserting agent, source relation, supporting observation count, rationale and provenance. Multiple evaluation records MAY concern the same pair of resources and MAY contain contradictory outcomes from different sources. The schema does not resolve such conflicts automatically.

The legacy property openrel:wasDerivedFrom accepts either an IRI or a literal source description. New records SHOULD prefer prov:wasDerivedFrom with an IRI, or dct:source for a textual citation.

A.4.5 Policy Comparison and Evaluation

A.4.5.1 Policy-Level Comparison

Two policies can be compared in the following ways:

1. The duties and obligations specified in a policy associated with an Asset (such as a openrel:Licence, an odrl:Assertion, openrel:AccessRules, openrel:ProcessRules) can be compared to a request for use of the Asset (odrl:Request/ openrel:RequestPolicy, odrl:Assertion/ openrel:AssertionPolicy), and on the basis of comparison of the duties and constraints in the two policies, it is possible to determine whether the request satisfies the provisions of the Asset policy or Policy Collection.
2. The actions and constraints defined in each policy can be compared to one another pairwise, and based on the outcome, compatibility (or not) of two policies can be determined. This approach is applicable in two specific sub-cases:
    a. Determining whether two policies can be applied to the same context, usually because the two policies are to be included in a PolicyCollection.
    b. Determining whether a derived Asset, based on separate policies applicable to two or more input Assets, can be combined in a Policy for the output Asset, or allows combination of the content of the two Assets[3].
3. Two policies can be compared on the basis of a signature, and determined to be exactly equivalent (or not).

Each of these comparison approaches are recorded utilising a special set of classes and properties defined by OpenREL, and not available in ODRL. The comparisons are supported in respect of

[3] Some of these cases are trivial: clearly, an asset licence that prohibits derivatives excludes combination of the asset content with that of another.

standardised outcome information by a concept scheme that predefines policy-level, action-level, and constraint-level outcomes.

The comparison of two policies uses three decision outcomes: PERMIT, CONDITIONAL and DENY.

Table A.4.5.1.1 openrel:typeEvaluationOutcome Concept Scheme: Policy-Level Comparison Outcomes

Outcome IRI	Decision	Required condition
openrel:policyEvaluation	-	Top-level concept. The remainder of the table defines the narrower concepts in the scheme.
openrel:policyPermit	PERMIT	All applicable prohibitions are accounted for and all preconditions, including preDuties, are satisfied. Post-access obligations may remain.
openrel:policyConditional	CONDITIONAL	No applicable prohibition exists, but one or more preDuties or preconditions remain unsatisfied.
openrel:policyDeny	DENY	An applicable prohibition exists, the policies cannot co-exist, or a mandatory condition cannot be satisfied. A structured, auditable reason is provided.

When a comparison cannot be resolved automatically, it requires human review. The eventual request decision still resolves to PERMIT, CONDITIONAL, or DENY.

PERMIT requires applicable prohibitions to be accounted for and preconditions, including PreDuties, to be satisfied; post-action obligations and PostDuties may remain. CONDITIONAL applies when no applicable prohibition exists but satisfiable preconditions remain outstanding. DENY applies when a prohibition exists or a mandatory condition cannot be satisfied; and an auditable reason is provided.

Policy and PolicyCollection comparisons may also draw on pairwise Action and Constraint comparisons. The aggregation algorithm and the treatment of unresolved comparisons are defined in the evaluation procedure, not by the RDF declarations in this section.

A.4.5.2 Pairwise Comparison of Actions

The primary use cases for this type of comparison are

1. To determine whether two policies can be applicable together.
2. To determine whether a derived work based on different policies for the Assets used in its composition is feasible.

The basis of comparison between actions is a knowledge base of mappings - the mappings contain predetermined outcomes of pairwise comparisons that are based on specific methods, as described in section 4.2. These methods of comparison (‘assertion’) are standardised and recorded for each comparison, and MUST correspond to one of the values defined in Table 4.5.2.1.

Concept scheme: `openrel:typeEvaluationOutcome`
Parent concept: `openrel:pairwiseEvaluation`

The following table lists the concepts used to describe the outcomes of pairwise comparisons between Actions or Constraints in OpenREL.

Table A.4.5.1.2 openrel:typeEvaluationOutcome Concept Scheme: Resource-Level Comparison Outcomes

Concept IRI	Preferred label	Definition	External mapping skos:seeAlso

openrel:pairwiseEvaluation	Pairwise Comparison Outcomes	Outcomes of pairwise comparisons between Actions or Constraints. The remainder of the table defines the narrower concepts in the scheme.	-
openrel:compatible	Action or Constraint is directly compatible with another	An Action or Constraint is directly compatible with another and both can be combined within the same Policy or PolicyCollection.	rdfs:seeAlso owl:sameAs
openrel:contradicts	Action or Constraint contradicts another	An Action or Constraint is incompatible with another and they cannot be combined within the same Policy or PolicyCollection.	skos:exactMatch dalicc:contradicts
openrel:allows	Allows	Indicates that a prohibition on the subject Action can coexist with a permission on the object Action within the same Policy.	-
openrel:includedIn	Action or Constraint is included in another	An Action or Constraint is included in another and they can be combined within the same Policy or PolicyCollection.	skos:closeMatch odrl:includedIn
openrel:implies	Action or Constraint implies another	An Action or Constraint implies another and they can be combined within the same Policy or PolicyCollection.	skos:closeMatch odrl:implies
openrel:missing	Action or Constraint is missing	An Action or Constraint is not present in the Policy being compared.	-

A.4.5.3 Consolidating Pairwise Comparisons

Annex B discusses the algorithm for comparing pairwise mappings of actions and constraints between two (or more) policies, and indicates the mechanisms for deriving a policy that satisfies both input policies.

A.4.6 Additional Conformance Specifications

OpenREL defines an optional SHACL validation module for checking the structural completeness, resource identification and provenance of evaluation records. The module supplements the RDF vocabulary definitions without imposing cardinality restrictions on the underlying RDF/OWL schema.

Implementations MAY apply the SHACL constraints to `openrel:EvaluationDetail`, `openrel:EvaluationMapping` and `openrel:PolicyCollection` instances.

SHACL validation results distinguish between:

- **`sh:Violation`** - a structural constraint corresponding to a mandatory conformance requirement.
- **`sh:Warning`** - a recommended practice that has not been satisfied.

A successful SHACL validation does not, by itself, establish the semantic correctness of an evaluation or the validity of a policy-level decision.

A.4.6.1 Evaluation Record Completeness

An openrel:EvaluationDetail SHOULD provide the properties in Table A.4.6.1

Table A.4.6.1 Evaluation Detail Property Constraints

Property	Constraint	Severity
openrel:comparedSubject	At least one value; IRI	sh:Warning
openrel:comparedIndividual	At least one value; IRI	sh:Warning
openrel:evaluationClass	At least one value; IRI	sh:Warning
openrel:evaluationOutcome	At least one value; IRI	sh:Warning
openrel:derivationType	At least one value; IRI	sh:Warning

These constraints are expressed using `sh:minCount 1` and `sh:nodeKind sh:IRI`.

Missing recommended properties produce warnings rather than violations. The validation module does not impose maximum cardinality restrictions.

A.4.6.2 Preservation of Compared Resource Identity

An OpenREL-conformant implementation MUST preserve the identity of the resources being compared.

Where `openrel:comparedSubject` and `openrel:comparedIndividual` are present, their values MUST be IRIs under the proposed SHACL validation profile.

The corresponding constraints use `sh:nodeKind sh:IRI` with severity `sh:Violation`.

A.4.6.3 Provenance and Rationale

An `openrel:EvaluationMapping` SHOULD preserve information identifying the source or responsible agent of an assertion.

The SHACL module issues a warning when neither of the following properties is present:

- `openrel:wasDerivedFrom` - identifies the source from which the assertion was derived.
- `openrel:assertedBy` - identifies the agent responsible for the assertion.

Implementations SHOULD additionally preserve a human-readable explanation using `openrel:rationale`, particularly for human-reviewed or manually asserted comparisons. The current SHACL module does not require `openrel:rationale`; its presence is a specification-level recommendation.

A.4.6.4 Policy-Level Evaluation References

An `openrel:PolicyCollection` MAY reference one or more `openrel:EvaluationDetail` instances using `openrel:hasEvaluationDetail`.

Every value of `openrel:hasEvaluationDetail` MUST identify an instance of `openrel:EvaluationDetail`.

This requirement is expressed using `sh:class openrel:EvaluationDetail` with severity `sh:Violation`.

The constraint validates the type of referenced evaluation records but does not require every PolicyCollection to have an evaluation record.

A.4.6.5 Assertion Classification and Policy Decision Integrity

An OpenREL-conformant implementation MUST distinguish explicitly asserted source relationships from inferred or derived comparisons.

The `openrel:derivationType` property SHOULD be used to record the derivation category, using concepts from `openrel:typeAssertion`.

An implementation MUST NOT treat an Action-level `openrel:EvaluationMapping` as sufficient evidence, by itself, for a final policy-level decision.

The overall policy decision MUST be determined through the applicable policy-evaluation procedure, considering relevant Permissions, Prohibitions, Duties, Constraints and supporting evidence.

These requirements are not fully enforceable by the structural SHACL constraints described here. They MUST be addressed by the evaluation implementation or by additional application-specific validation procedures.

A.4.6.6 Conformance Summary

Table A.4.6.1 Evaluation Conformance Summary

Requirement	Normative level	SHACL mechanism
Identify the five recommended evaluation properties	SHOULD	sh:minCount 1, sh:Warning
Preserve the identity of compared resources	MUST	sh:nodeKind, sh:Violation
Preserve assertion provenance	SHOULD	sh:or, sh:Warning
Preserve human-review rationale	SHOULD	Specification-level recommendation
Validate PolicyCollection evaluation references	MUST	sh:class, sh:Violation
Distinguish asserted from inferred comparisons	MUST	Implementation-level validation
Prevent unsupported policy-level decisions	MUST NOT	Evaluation algorithm / application validation

The optional SHACL module provides structural conformance checking. It does not replace the OpenREL policy-evaluation algorithm or establish the semantic validity of comparison outcomes.

A.5. Constraints and Named Constraints

A.5.1 Scope and Rationale

ODRL defines the odrl:Constraint class as a mechanism for encoding the operands and operators involved in definition of a constraint, and also allows logical constraints to be created from a combination of two or more constraint nodes. The constraint nodes, after definition, are considered ‘local’ to a policy, and there is no explicit provision for reuse of constraints across different policies, although this is technically possible.

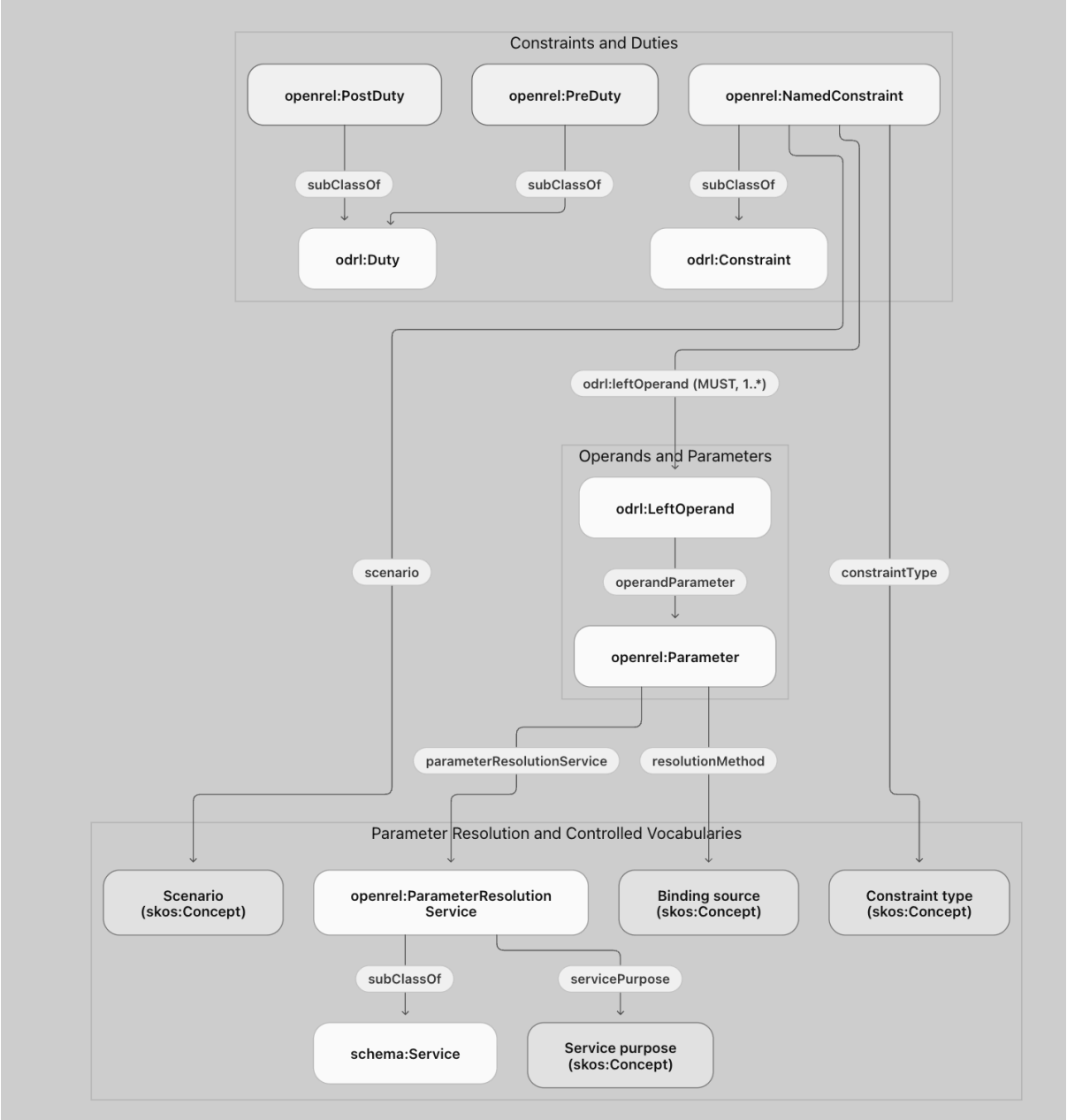
OpenREL defines openrel:NamedConstraint as a reusable, individually identified subclass of odrl:Constraint. Named Constraints MAY be published in a registry and referenced by multiple policies or rules, allowing consistent reuse of standard constraint expressions. They retain the ODRL constraint structure: odrl:leftOperand, odrl:operator and odrl:rightOperand. The Named Constraint

model adds classification and contextual metadata; it does not change the semantics of these ODRL properties.

A Named Constraint is an RDF resource with a stable IRI. A registry SHOULD provide a label, a definition and the applicable ODRL expression. The registry MAY also provide the additional metadata defined below. Whether a Named Constraint is satisfied depends on the evaluation context, the relevant operands and any required parameter bindings.

Figure A.5.1 presents the information model for Parameters, Named Constraints, Left Operands, and related concepts.

Figure A.5.1: Information Model - Parameters, Named Constraints, and Left Operands



A.5.2 Class Definition

Table A.5.2 Named Constraint Subclass

Class	Superclass	Definition
openrel:NamedConstraint	odrl:Constraint	A reusable, individually identified ODRL Constraint that may be published in a registry and referenced by policies or rules.

The schema declares openrel:NamedConstraint as owl:Class and skos:Concept, following the OpenREL documentation convention. It does not redeclare or redefine odrl:Constraint.

A.5.2.1 Named Constraint

IRI: openrel:NamedConstraint | Superclass: odrl:Constraint

A reusable, individually identified ODRL Constraint that may be published in a registry and referenced by policies or rules. Named Constraints use the ODRL leftOperand, operator and rightOperand structure. Their additional OpenREL properties support classification, scenarios, legal basis and relevant endpoints.

A.5.3 Additional Properties

Table A.5.3 Named Constraint Properties

Property	Domain / Range	Purpose
openrel:constraintScheme (Concept Scheme)	-	OpenREL Named Constraint class and property definitions are included in a concept scheme.
openrel:namedConstraint	Action / NamedConstraint	Optional link to a reusable Named Constraint for a Rule containing an Action.
openrel:applicableEndpoint	NamedConstraint / resource IRI	Endpoint used for notification, specification or evaluation.This can be an automated test.
openrel:hasLegalBasis	NamedConstraint or Rule/ resource IRI	Relevant legal basis, potentially expressed using DPV.
openrel:constraintType	NamedConstraint / skos:Concept	Classification by purpose or rationale.
openrel:scenario	NamedConstraint / skos:Concept	Scenario or context to which the constraint applies.

The intended alternative domains for namedConstraint and hasLegalBasis are documented in prose rather than declared as global rdfs:domain union restrictions. applicableEndpoint uses resource IRIs, rather than the literal datatype xsd:anyURI.

Values of openrel:hasLegalBasis SHOULD be populated using IRIs provided by the DPV Legal Basis vocabulary items, these can be found here: https://dev.dpvcg.org/dpv/modules/legal_basis. The value MAY, however, also be populated with a literal or from any other relevant vocabulary if the legal basis for a constraint is not derived from privacy legislation such as GDPR.

A.5.4 Expression and Reuse

Each Named Constraint SHOULD provide the ODRL properties odrl:leftOperand, odrl:operator and odrl:rightOperand. The Named Constraint MAY be associated with a Policy or Rule through openrel:namedConstraint. Standard ODRL links and interpretation remain applicable.

```
@prefix openrel: <http://www.w3id.org/ns/openrel/0/> .
@prefix odrl: <http://www.w3.org/ns/odrl/2/> .
```

```
openrel:academicUseOnly a openrel:NamedConstraint ;
  odrl:leftOperand openrel:usePurpose ;
  odrl:operator odrl:eq ;
  odrl:rightOperand openrel:academicResearch ;
  openrel:constraintType openrel:purpose .
```

Named constraints Left Operands make extensive use of predefined concept schemes to support consistency and reusability, and many Right Operands are based on context-dependent parameters that are also predefined for the same reason. These links are discussed in detail in the relevant sections.

A.5.5 Registry and Evaluation Considerations

The current registry contains 27 Named Constraint individuals. A policy evaluator SHOULD resolve a referenced Named Constraint to its ODRL expression and evaluate it using the applicable context. Publication in a registry does not, by itself, establish satisfaction of a constraint or authorise an Action.

Named Constraints MAY be categorised by constraintType and Scenario. The semantics and allowed values of these properties are supplied by separately maintained concept schemes. An evaluator MUST NOT assume that an undefined LeftOperand or right-operand concept has a universally understood evaluation procedure.

The latest authoritative registry defines 27 Named Constraint individuals. The table below lists their published identifiers and definitions.

A.5.5.1 Named Constraint Registry

Registry IRI: openrel:NamedConstraintRegistry

Table A.5.5.1 Named Constraint Registry Individuals

Named Constraint IRI	Preferred label	Definition	Constraint Type
openrel:academicUseOnly	academicUseOnly	Asset MUST only be used for Academic Research or Applications	purpose
openrel:allowDistributionFee	allowDistributionFee	Allow legitimate and reasonable costs of distribution of the asset or its derivatives without treating these costs as commercial use.	revenue
openrel:calculatedStartDate	calculatedStartDate	Permits the associated action only when the evaluation date and time is on or after a dynamically calculated start date.	-
openrel:confirmedAIAgent	confirmedAIAgent	The agent has been verified as such.	context
openrel:consentLegal	consentLegal	Allow reuse of the Asset or Asset Collection where consent provides the applicable legal basis.	consent
openrel:consentPrior	consentPrior	Allow access to the Asset or Asset Collection only when applicable consent	consent

		has been obtained before access, and this has been verified.	
openrel:countLimit	countLimit	A usage constraint requiring the number of executions of an Action within a defined accounting scope to remain within a specified maximum.	usage
openrel:distributeUnmodified	distributeUnmodified	Distribution is permitted only when the distributed material is not an adaptation of the licensed material.	-
openrel:eidasVerified	eidasVerified	The end user or agent has been verified via eIDAS.	context
openrel:eidasVerifiedResearcher	eidasVerifiedResearcher	The requestor has an eIDAS-verified identity and an accepted researcher-status assertion associated with that identity.	context
openrel:ethicsApproval	ethicsApproval	Allow access to the Asset or Asset Collection only when an institutional review board has confirmed ethics approval and this has been verified.	consent
openrel:euUseOnly	euUseOnly	Allow the Asset to be used within the current definition of the EU.	context
openrel:externalRecipient	externalRecipient	Identifies a distribution request where the intended recipient is not affiliated with the designated institution.	-
openrel:institutionalAffiliation	institutionalAffiliation	Requires the evaluated Assignee to have verified affiliation with the designated institution.	-
openrel:institutionalInfrastructure	institutionalInfrastructure	Requires an Asset to be used only on infrastructure authorised by the designated institution.	-
openrel:logUse	logUse	Instances of use of the asset or asset collection must be recorded in a log and retained for a legally mandated duration.	notification
openrel:managedAccess	managedAccess	Automated access only permitted for managed access policies.	tom
openrel:noSpecialCategory	noSpecialCategory	Consent does not include legitimate use as a special data category (GDPR Art 9.2 (a)).	consent

openrel:notifyCitation	notifyCitation	Use of the asset must be acknowledged through appropriate citation in derived works and assets.	notification
openrel:notifyClearingHouse	notifyClearingHouse	Instances of use of the asset or asset collection must be notified to a clearinghouse endpoint.	notification
openrel:notifyInAsset	notifyInAsset	Changes and modifications must be added in the Asset using human-readable text, without removing the current licence information.	notification
openrel:openAccess	openAccess	Automated access only permitted for open access policies.	tom
openrel:trustPlus	trustPlus	Automated access - inclusive of AI agents - must be verified against the TRUST+ specification.	access
openrel:validEndDate	validEndDate	Access is only permitted on or before the specified end date.	tom
openrel:validStartDate	validStartDate	Access is only permitted on or after the specified start date.	tom
openrel:verifiedResearcher	verifiedResearcher	A constraint requiring the evaluated agent to have a recognised researcher role, established through an accepted assertion or verification mechanism.	role
openrel:volumeLimit	volumeLimit	A usage constraint requiring the measured or projected cumulative data volume associated with an Action to remain within a specified maximum.	usage

A.5.6 Applicable Concept Schemes

The values for openrel:constraintType MUST be populated from the concept scheme defined below.

Concept Scheme IRI: openrel:typeConstraint

Table A.5.6 openrel:typeConstraint Concept Scheme

Concept IRI	Preferred label	Definition	Related vocabulary
openrel:role	Role	A Named Constraint based on the role of an Agent.	type_agent
openrel:usage	Usage	A Named Constraint based on characteristics of asset usage.	type_usage
openrel:revenue	Revenue	A Named Constraint based on permitted or prohibited revenue generation.	type_revenue

openrel:access	Access	A Named Constraint based on permitted access types or conditions.	type_access
openrel:purpose	Purpose	A Named Constraint based on permitted purposes of use, including research purposes.	type_purpose
openrel:notification	Notification	A Named Constraint specifying conditions associated with notification duties.	type_notification-target
openrel:consent	Consent	A Named Constraint specifying consent-related conditions or requirements.	type_consent
openrel:context	Context	A Named Constraint based on contextual characteristics of an Agent or its operating environment.	type_context
openrel:tom	Technical and Organisational Measures	A Named Constraint specifying technical, organisational, legal or physical measures required to safeguard sensitive data.	type_tom, DPV TOM

OpenREL defines the SKOS ConceptScheme openrel:typeConstraint to classify Named Constraints according to their intended purpose and the characteristics or conditions they represent.

Each openrel:NamedConstraint MAY be associated with one or more constraint types through the openrel:constraintType property. The values SHOULD be selected from the openrel:typeConstraint concept scheme.

The scheme defines eleven categories: Agent Role, Usage, Revenue, Access, Purpose, Notification, Consent, Context, Subject Rights, Technical and Organisational Measures (TOM), and Trust Connection Types.

These categories support the organisation, discovery and reuse of Named Constraints in published registries. They also facilitate the identification of relevant controlled vocabularies for interpreting constraint operands and values.

The categories are not mutually exclusive. A Named Constraint MAY be associated with multiple categories where its application spans more than one type of condition. The classification does not change the semantics of the underlying ODRL Constraint. The interpretation of a Named Constraint remains determined by its operands, operator, associated parameters and applicable policy context.

Where appropriate, OpenREL reuses external vocabularies, including the Data Privacy Vocabulary (DPV), to describe technical, organisational and other safeguarding measures.

Each of these high-level constraint types is optionally supported by a more detailed concept scheme. These are presented in the sections that follow.

A.5.6.1 Agent Type

OpenREL defines the SKOS ConceptScheme `openrel:typeAgent` to provide a controlled vocabulary for identifying types and roles of agents involved in accessing, using, managing or otherwise interacting with Assets governed by ODRL and OpenREL Policies.

The vocabulary supports the specification of Named Constraints, particularly those classified as role-based or context-based constraints. For example, the `openrel:agentRole` LeftOperand MAY

reference concepts from this scheme to identify the role or type of an agent participating in a policy evaluation.

The scheme defines six concepts: AI Agent, Human Agent, Research Organisation, Researcher, Data Subject and Curator.

Concept Scheme IRI: openrel:typeAgent

Table A.5.6.1.1 openrel:typeAgent Concept Scheme

Concept IRI	Preferred label	Reviewed definition	Modelling note
openrel:ai-agent	AI Agent	An AI-based software agent or autonomous process that seeks to access or use an Asset.	Agent type
openrel:human-agent	Human Agent	A human individual who seeks to access or use an Asset.	Agent type
openrel:research-organisation	Research Organisation	An organisation whose status as a research-performing or academic institution can be independently verified.	Organisation type
openrel:researcher	Researcher	An agent whose status or role as a researcher can be independently verified.	Agent role
openrel:data-subject	Data Subject	A person, legal entity, community, ecosystem or geographic region that is the subject of research or information represented by an Asset.	Subject relationship; broader than the GDPR definition
openrel:curator	Curator	An agent responsible for the curation, publication, archiving, dissemination or preservation of an Asset.	Agent role

These concepts represent different aspects of agent classification. AI Agent and Human Agent distinguish types of actors; Research Organisation identifies an institutional category; Researcher and Curator describe roles or responsibilities. The Data Subject concept identifies an entity about which information is collected or represented and is not necessarily an agent participating in a policy transaction.

An agent MAY have multiple classifications. For example, a human agent MAY also be a researcher and a curator. An example encoding may look like this:

```
openrel:verifiedResearcher
   a openrel:NamedConstraint ;
   odrl:leftOperand openrel:agentRole ;
   odrl:operator odrl:eq ;
   odrl:rightOperand openrel:researcher .
```

Where an OpenREL Policy requires verification of an agent's role or status, the applicable Named Constraint SHOULD identify the relevant classification and MAY reference additional evidence or verification mechanisms.

The classification of an agent as, for example, a Researcher or Research Organisation does not, by itself, establish that the agent has been independently verified. Verification requirements and evidence or assertions of evidence are defined separately.

Implementations MAY map OpenREL agent concepts to external vocabularies, including FOAF, DCAT-AP and DPV, where appropriate. Such mappings MUST distinguish between concept-level equivalence and relationships to external RDF classes or properties. A mapping to an external class MUST NOT be interpreted as SKOS concept equivalence unless that interpretation is explicitly justified.

The OpenREL Data Subject concept has a broader scope than the GDPR definition, encompassing persons, organisations, communities, ecosystems and geographic regions. Implementations dealing with personal data SHOULD use the relevant DPV terminology where the narrower legal meaning is required.

An implementation using the OpenREL agent classification vocabulary SHOULD:

- Use the published concept IRIs when referencing the defined agent types and roles.
- Permit multiple classifications for an individual agent.
- Distinguish a declared role from a verified role.
- Apply additional verification requirements where a Policy or Named Constraint requires evidence of agent status.
- Avoid inferring legal rights or permissions solely from an agent classification.

The classification vocabulary supports policy evaluation but does not independently grant or restrict access to an Asset.

ODRL makes provision for the definition of a Party or PartyCollection to unambiguously identify the part (agent) involved either as a rights holder or requestor in respect of an Asset. The context of OpenREL Actor types and ODRL Party and PartyCollection objects is shown below.

Table A.5.6.1.2 Role of Party, PartyCollection, and Agents

Concept	**Purpose**	**Typical representation**
odrl:Party	Identifies an entity participating in a policy, such as an assigner or assignee.	An individual resource with an IRI
odrl:PartyCollection	Identifies a collection of Parties, potentially with membership or refinement conditions.	An individual resource with an IRI
openrel:typeAgent	SKOS concept scheme classifying agent types and roles.	A controlled vocabulary
openrel:researcher, openrel:curator, etc.	Classification values used when evaluating an agent's characteristics or role.	skos:Concept individuals

A.5.6.2 Type of Use

OpenREL defines the SKOS ConceptScheme openrel:typeUsage to classify usage conditions that may be applied to Actions through ODRL Constraints and OpenREL Named Constraints.

The scheme provides a controlled vocabulary for describing common usage-related restrictions on access, processing and use of Assets. These include restrictions based on connectors, applications, IP addresses, duration, temporal intervals, geographic locations, user roles, purposes, usage counts, data volumes and security assurance levels.

Usage Types are classification concepts and MUST NOT be interpreted as ODRL Constraints or LeftOperands solely by virtue of their membership in this scheme.

Concept Scheme IRI: openrel:typeUsage

Table A.5.6.2 openrel:typeUsage Concept Scheme

Concept IRI	Preferred label	Definition	Related IDS policy class
openrel:connector	Connector	The identifier of the data connector through which the evaluated Action is requested or performed.	idsc:CONNECTOR_RESTRICTED_DATA_USAGE See also 5.6.11
openrel:specificIpRange	Specific IP Range	The IP address of the network endpoint associated with the evaluated Action.	-
openrel:application	Application	The identifier of the application through which the evaluated Action is requested or performed.	idsc:APPLICATION_RESTRICTED_DATA_USAGE
openrel:accessDuration	Access Duration	The elapsed duration of access measured from a defined starting event to the evaluation time or proposed end of access.	idsc:DURATION_RESTRICTED_DATA_USAGE
openrel:accessWindow	Access Window	A usage condition restricting access to a specified date-time interval.	idsc:INTERVAL_RESTRICTED_DATA_USAGE
openrel:spatialCoverage	Spatial Coverage	The geographic location or spatial extent associated with the evaluated Action, agent or system.	idsc:LOCATION_RESTRICTED_DATA_USAGE
openrel:userRole	User Role	The role held by the relevant Party in the evaluated Action.	idsc:ROLE_RESTRICTED_DATA_USAGE
openrel:usagePurpose	Usage Purpose	The declared or verified purpose associated with the evaluated Action.	idsc:PURPOSE_RESTRICTED_DATA_USAGE
openrel:usageCount	Usage Count	The accumulated number of executions of the evaluated Action within the applicable accounting scope, including the proposed execution.	idsc:RESTRICTED_NUMBER_OF_USAGES
openrel:usageVolume	Usage Volume	The quantity of data associated with executions of an Action, measured within a defined accounting scope.	-

openrel:securityLevel	Security Level	The security assurance level established for the connector or system associated with the evaluated Action.	idsc:SECURITY_LEVEL_RESTRICTED_POLICY
openrel:usageEvent	Usage Event	The event identifier or event type associated with the evaluated Action.	idsc:EVENT_RESTRICTED_DATA_USAGE

A Usage Type identifies the nature of a usage condition. The corresponding executable restriction is represented by an ODRL Constraint or an OpenREL Named Constraint.

The Constraint specifies the applicable `odrl:leftOperand`, `odrl:operator` and `odrl:rightOperand`, together with any additional properties or parameters required for evaluation.

For example, a duration-based restriction may use a parameter of datatype `xsd:duration`, while an access-window restriction may require two date-time values and a logical conjunction of boundary conditions.

The precise representation and evaluation of these values are addressed in the OpenREL LeftOperands and Parameters sections.

Several OpenREL Usage Types correspond to usage-policy patterns defined by the International Data Spaces information model. OpenREL maintains its own concept IRIs to provide a stable classification vocabulary independent of external policy-class definitions. Where a related IDS Usage Policy Class exists, the OpenREL concept MAY reference it using `rdfs:seeAlso`. Such references are informative and do not assert semantic equivalence, class inheritance or interchangeability.

Implementations MAY use external vocabularies to define specific operand values, units of measure, locations, security assurance levels and other evaluation parameters.

Some Usage Types require parameterised values that are supplied during policy authoring or evaluation. Examples include connector identifiers, duration limits, date-time boundaries, geographic regions, maximum usage counts and permitted data volumes.

Implementations MUST evaluate the relevant values using the operator and comparison semantics specified by the applicable Constraint.

Where evaluation requires contextual information, such as the identity of a connector, the current date and time, or the number of previous access events, the implementation MUST obtain this information from an appropriate evaluation context or authoritative source.

A Usage Type classification does not, by itself, grant or prohibit an Action.

Implementations using the OpenREL Usage Types vocabulary SHOULD use the published concept IRIs and SHOULD preserve the distinction between classification concepts, ODRL LeftOperands and Named Constraint instances.

Implementations MUST NOT infer policy permissions or prohibitions solely from membership in `openrel:typeUsage`.

Implementations MAY extend the scheme with additional concepts, provided that their definitions and relationships to existing concepts are documented.

A.5.6.3 Revenue Type

The Revenue Source Types vocabulary defines a SKOS ConceptScheme, trsp:revenue-source-type-scheme, for classifying revenue sources, funding arrangements and charging mechanisms associated with the provision and use of assets and services. OpenREL does not directly specify revenue types, but work being done in the EOSC EDEN/ FIDELIS projects can be considered as a basis for unambiguous identification of revenue types or sources.

Concept Scheme IRI: trsp:revenue-source-type-scheme

Table A.5.6 trsp:revenue-source-type-scheme Concept Scheme

Concept IRI	Preferred label	Definition
`trsp:free`	Free	No monetary charge is imposed on the user for specified access, use or service.
`trsp:subscription`	Subscription	Recurring payment provides access to an asset or service for a defined period.
`trsp:payPerUse`	Pay per Use	Payment is determined by the number or quantity of uses of an asset or service.
`trsp:licenceFee`	Licence Fee	Monetary charge associated with obtaining or exercising rights granted under a licence.
`trsp:distributionCost`	Recovery of Distribution Costs	Charges intended to recover costs directly associated with distributing or delivering an asset.
`trsp:structural`	Structural Income or Budget Allocation	Recurring institutional funding or budget allocations supporting an asset or service.
`trsp:membership`	Membership	Funding or charging based on membership contributions or fees.
`trsp:grants`	Grants	Funding provided through grants or similar financial awards.

The scheme includes eight concepts: Free, Subscription, Pay per Use, Licence Fee, Recovery of Distribution Costs, Structural Income or Budget Allocation, Membership, and Grants. The concepts distinguish between arrangements in which charges are imposed directly on users and arrangements through which assets or services receive institutional or external funding.

OpenREL MAY reference Revenue Source Types when describing financial conditions associated with the use, distribution or provision of an Asset.

In particular, a Named Constraint classified under openrel:revenue MAY reference a Revenue Source Type to identify the applicable charging or funding arrangement that is permitted.

The Revenue Source Type provides a classification and does not independently specify a monetary amount, payment frequency, currency, recipient or contractual obligation. ODRL allows specification of these amounts by way of constraint expressions.

For example, a restriction permitting only the recovery of distribution costs MAY reference trsp:distributionCost. The actual financial threshold and the conditions governing cost recovery must be specified separately.

The concepts are not necessarily mutually exclusive. An asset or service may be supported through institutional budget allocations and grants while also applying subscription or pay-per-use charges.

Implementations SHOULD distinguish revenue sources from charging mechanisms when interpreting or aggregating classifications.

A.5.6.4 Access Type

This makes use of the same classification as presented in A.1.8.1.

A.5.6.6 Purpose

The purpose for which access to an Asset is requested, or a specific constraint on purpose of use can be specified or constrained using this concept scheme. OpenREL defines the SKOS ConceptScheme `openrel:type-purpose` to provide a controlled vocabulary for classifying the purposes for which Assets may be accessed, used or processed.

Concept Scheme IRI: openrel:type-purpose

Table A.5.6.6 openrel:type-purpose Concept Scheme

Concept IRI	Preferred label	Definition	External reference
openrel:researchUse	General Research Use	Use of an Asset for research purposes without restriction to a particular research discipline, field or application.	DUO:0000042; DPV ResearchAndDevelopment
openrel:academicResearch	Academic Research	Research conducted in an academic context, including research undertaken within universities and other academic institutions.	DPV AcademicResearch
openrel:commercialResearch	Commercial Research	Research conducted in a commercial context, including research undertaken by or for commercial organisations.	DPV CommercialResearch
openrel:nonCommercialResearch	Non-Commercial Research	Research undertaken for purposes that do not constitute commercial use under the applicable policy or licence.	DPV NonCommercialResearch
openrel:specificDiscipline	Specific Discipline Research	Research undertaken within one or more specified scientific disciplines or fields of study.	DUO:0000012
openrel:clinicalCareUse	Clinical Care Use	Use of an Asset for clinical care, healthcare delivery or clinical decision-making.	DUO:0000043
openrel:healthBiomedicalResearch	Health, Medical or Biomedical Research	Research undertaken for health, medical or biomedical purposes.	DUO:0000006
openrel:geneticStudies	Genetic Studies	Research involving genetic information or genotypes, including studies combining	DUO:0000016

		genotype and phenotype information.	
openrel:diseaseSpecific	Disease-Specific Research	Research directed towards one or more specified diseases or disease categories.	DUO:0000007
openrel:populationResearch	Population Origins or Ancestry Research	Research investigating population origins, ancestry or related population histories.	DUO:0000011; DUO:0000044
openrel:generalMethods	General Methods Research	Research directed towards developing or evaluating general methods, including software, algorithms and analytical techniques.	DUO:0000015

The vocabulary includes general research purposes, academic and commercial research, discipline-specific research, clinical care, biomedical research, genetic studies, disease-specific research, population-origins research and methods development.

The concepts are intended to describe purposes independently of whether a particular purpose is permitted, prohibited or subject to additional conditions.

Purpose Types MAY be used as right-operand values in ODRL Constraints and OpenREL Named Constraints. A Purpose Type identifies the purpose being evaluated. Whether that purpose is permitted or prohibited is determined by the applicable ODRL Rule and its associated Constraints. For example, a Permission MAY restrict use of an Asset to academic research, while a Prohibition MAY exclude use for commercial research.

OpenREL Purpose Types are informed by the Data Use Ontology (DUO) and the Data Privacy Vocabulary (DPV). DUO defines standardised data-use permissions and modifiers, particularly for biomedical research datasets. DPV defines purposes relevant to data processing and privacy governance. OpenREL maintains separate purpose concepts to support the expression of both Permissions and Prohibitions without embedding the direction of the restriction in the purpose classification, but implementers MAY choose to use the DPV and/ or DUO vocabularies directly.

Certain purposes require additional information to support policy evaluation. For example:

- Specific Discipline Research requires identification of the applicable discipline.
- Disease-Specific Research requires identification of the relevant disease or disease category.
- Genetic Studies may require clarification of the permitted study types.
- Non-Commercial Research requires an applicable interpretation of commercial activity.

Such additional information SHOULD be represented using appropriate ODRL Constraints, OpenREL Parameters or external controlled vocabularies.

Implementations MAY associate multiple purposes with an Action, subject to the logical operators and conditions specified by the applicable policy.

A.5.6.7 Notification Type

OpenREL defines the SKOS ConceptScheme `openrel:notification-target-type` to classify notification, logging, recording and communication mechanisms associated with the access, use, processing and management of Assets.

Concept Scheme IRI: openrel:notification-target-type

Table A.5.6.7 openrel:notification-target-type Concept Scheme

Concept IRI	Preferred label	Definition	External reference
openrel:notify-attribution	Attribution Notification	A notification or record identifying the attribution or citation associated with the use of an Asset.	-
openrel:notify-clearinghouse	Clearinghouse Notification	A notification of Asset access or use submitted to a designated clearinghouse or audit service.	IDS Remote Notification
openrel:notify-external-log	External Logging Notification	A notification or usage record submitted to an external logging or monitoring service.	IDS Remote Notification
openrel:notify-local-log	Local Logging	The recording of specified Asset access, use or processing events in a local log.	IDS Local Logging
openrel:notify-ropa	Record of Processing Activities (ROPA)	The documentation of relevant personal-data processing activities in a Record of Processing Activities.	GDPR Article 30; DPV ROPA
openrel:record-in-asset	Record in Asset	The inclusion of a notice or record within an Asset documenting its use, modification or other relevant actions.	DALICC modificationNotice
openrel:notice-collection-subject	Direct Collection Notice to Data Subject	Information provided to a data subject when personal data are collected directly from that individual.	GDPR Article 13
openrel:notice-collection-indirect	Indirect Collection Notice to Data Subject	Information provided to a data subject when personal data have been obtained from a source other than that individual.	GDPR Article 14
openrel:notice-actions	Notification of Rights-Related Actions	A communication concerning actions taken in response to the exercise of data subject rights, including rectification, erasure or restriction of processing.	GDPR Article 19
openrel:notice-sar	Subject Access Request Response	A response providing information or access to personal data following the exercise of a data subject's right of access.	GDPR Article 15

- Implementations SHOULD use the published OpenREL concept IRIs when classifying notification and recording requirements.
- Implementations MUST distinguish a notification type from the corresponding Duty, recipient, destination and evidence of fulfilment.
- Where a notification or recording obligation is mandatory, the applicable policy MUST specify sufficient information for the requirement to be evaluated.
- Implementations MAY associate notification types with additional constraints, parameters or external vocabularies.

The vocabulary supports the representation of notification-related requirements in OpenREL policies, including requirements for attribution, reporting to external services, local logging, maintaining processing records, embedding notices in Assets, and communicating with data subjects.

Notification and recording requirements are generally expressed through ODRL Duties. An applicable Duty MAY require an assignee or another designated Party to submit a notification, maintain a record, or communicate information to an identified recipient.

OpenREL Named Constraints MAY be used to specify additional conditions associated with such Duties, including the notification type, recipient, destination, content, format, timing or other applicable requirements. The Notification Type concept identifies the nature of the required activity. It does not independently create an obligation or determine whether the activity has been completed.

The vocabulary distinguishes between notifications transmitted to recipients or services and information recorded in a designated location. External notification mechanisms include clearinghouse notifications and notifications submitted to external logging services. Local recording mechanisms include local logging, maintaining records of processing activities and recording notices within an Asset.

A notification or recording requirement MAY specify an endpoint, recipient, record location or other destination. Such destinations SHOULD be represented separately from the classification concept, using an IRI or another appropriate identifier and utilising the openrel:applicableEndpoint property.

OpenREL includes notification types related to the information and communication obligations established under the General Data Protection Regulation. These include notices concerning direct and indirect collection of personal data, responses to subject access requests, and communications associated with the exercise of data subject rights. The applicable GDPR Articles are referenced as sources for the corresponding notification types. The OpenREL concept of a Data Subject may encompass entities beyond natural persons. However, GDPR-specific notification obligations apply to personal data concerning natural persons within the scope of that Regulation.

OpenREL Notification Types may reference related policy patterns and concepts from the International Data Spaces information model, DALICC and the Data Privacy Vocabulary. These references are informative unless a formally justified semantic mapping has been established. In particular, an external usage-policy class describing a notification obligation is not necessarily equivalent to an OpenREL concept classifying a notification mechanism.

A.5.6.8 Consent, Approval, and Authorisation Types

OpenREL defines the SKOS ConceptScheme `openrel:typeConsent` to classify consent, approval, clearance and authorisation requirements associated with the use, processing, reuse, modification and dissemination of Assets.

The vocabulary supports requirements arising from legal, ethical, contractual and community-governance frameworks.

Concept Scheme IRI: openrel:typeConsent

Table A.5.6.8 openrel:typeConsent Concept Scheme

Concept IRI	Preferred label	Definition	External reference
openrel:consent-ethics	Ethics Clearance	A requirement for documented approval or clearance from an authorised research ethics committee or institutional review board before undertaking a specified activity.	-

openrel:consent-informed	Informed Consent	A requirement to obtain agreement from an appropriately informed and authorised person or representative before undertaking a specified activity involving an Asset or protected interest.	-
openrel:consent-legal	GDPR Consent	A requirement for valid consent from the data subject where consent is relied upon as the lawful basis for processing personal data under the GDPR.	dpv:Consent; GDPR Articles 4, 6, 7
openrel:consent-reuse-before	Prior Reuse Consent	A requirement to obtain consent or approval from an authorised subject, rights holder, creator or curator before undertaking a specified reuse of an Asset.	-
openrel:consent-reuse-after	Post-Reuse Dissemination Consent	A requirement to obtain consent or approval from an authorised Party before publishing, distributing or otherwise disseminating an Asset derived or modified through an earlier reuse activity.	-
openrel:approval-community	Community Approval	A requirement to obtain approval from an appropriately authorised community or its representatives before using, reusing, publishing or disseminating an Asset or information affecting that community's interests.	CARE Principles
openrel:authorisation-contractual	Contractual Authorisation	A requirement to obtain authorisation under an applicable contract or agreement before using, reusing, publishing or disseminating an Asset.	-

The scheme includes seven concepts: Ethics Clearance, Informed Consent, GDPR Consent, Prior Reuse Consent, Post-Reuse Dissemination Consent, Community Approval and Contractual Authorisation.

- Consent and authorisation requirements MAY be represented through ODRL Constraints, Duties and OpenREL Named Constraints.
- A Constraint MAY require evidence that a specified type of consent or authorisation has been obtained before an Action is permitted.
- An ODRL Duty MAY require a Party to request or obtain an approval, where the applicable policy defines such an obligation.
- The relevant policy MUST distinguish the type of required consent or authorisation from the Party responsible for providing it and the evidence demonstrating its existence.
- A Consent Type MUST NOT be interpreted as independently granting permission to perform an Action.
- Implementations applying GDPR consent MUST consider the applicable legal requirements, including the validity, specificity, scope and withdrawal of consent.
- Implementations SHOULD use the published OpenREL concept IRIs to classify consent and authorisation requirements.
- Implementations MUST distinguish the requirement for consent from the evidence that consent has been obtained.

- Where consent or authorisation is a prerequisite for an Action, the applicable policy MUST provide sufficient information to determine whether the requirement is satisfied.
- Implementations MAY associate Consent Types with additional constraints, parameters and external vocabularies.

The vocabulary distinguishes informed consent from consent recognised under specific legal frameworks. `openrel:consent-informed` represents a general requirement for informed agreement from an appropriately authorised person or representative. `openrel:consent-legal` identifies consent relied upon as a lawful basis for personal-data processing under the GDPR. The GDPR concept is related to `dpv:Consent` and to the relevant provisions of Regulation (EU) 2016/679.

Ethics clearance is represented separately because approval by an ethics committee or institutional review board does not necessarily replace consent from research participants or other affected parties.

OpenREL supports authorisation requirements relating to interests that extend beyond personal data. Examples include indigenous knowledge, community-held cultural information, commercially sensitive information and data concerning threatened species or ecosystems.

Consent and authorisation requirements may apply at different stages of an Asset's lifecycle. Prior Reuse Consent applies before a specified reuse activity. Post-Reuse Dissemination Consent applies before publication or dissemination of an Asset produced through earlier reuse.

A.5.6.9 End User Context

OpenREL defines the SKOS ConceptScheme openrel:typeContext to classify sources and mechanisms that may provide evidence concerning the identity, affiliation, research activity, organisational status or other relevant characteristics of a requestor. The scheme supports the evaluation of context-dependent policy conditions, particularly where access or use of an Asset depends on the verified context of the requesting Party.

Concept Scheme IRI: openrel:typeContext

Table A.5.6.9 openrel:typeContext Concept Scheme

Concept IRI	Preferred label	Definition	External reference
openrel:context-eidas	eIDAS Verification	Verification of identity or specified attributes using electronic identification or attestations supported by the eIDAS framework.	eIDAS
openrel:context-orcid	ORCID Verification	Verification of an ORCID identifier or associated assertions as evidence relevant to the identity, affiliation or research activity of a requestor.	ORCID
openrel:context-edugain	eduGAIN Verification	Verification of authentication or identity-provider attributes obtained through an identity federation participating in eduGAIN.	eduGAIN
openrel:context-skg	Scholarly Knowledge Graph Verification	Verification of claims concerning research activity, scholarly outputs or affiliations using records from a scholarly knowledge graph.	OpenAIRE Graph, OpenAlex DataCite PID Graph

openrel:context-cordis	CORDIS Verification	Verification of an organisation's participation in an EU-funded research or innovation project using CORDIS records.	CORDIS
openrel:context-ror	ROR Verification	Verification of an organisation's identity or registry attributes using the Research Organization Registry.	ROR

A.5.6.10 Technical and Organisational Measures

OpenREL defines the SKOS ConceptScheme `openrel:type-tom` to classify additional technical and organisational measures used to safeguard Assets and regulate their access, processing, use and disclosure.

The scheme complements existing vocabularies, particularly the Data Privacy Vocabulary (DPV), by providing concepts for operational safeguards relevant to sensitive research data and other protected information.

Concept Scheme IRI: openrel:type-tom

Table A.5.6.10 openrel:type-tom Concept Scheme

Concept IRI	Preferred label	Definition	Related vocabulary
openrel:limitResolution	Spatial Resolution Limitation	A measure that limits the spatial detail of information made available through an Asset or its derived outputs.	dpv:TechnicalMeasure
openrel:dataMinimisation	Data Minimisation	A measure that limits the data made available or processed to the subset necessary for a specified purpose or authorised activity.	dpv:TechnicalMeasure
openrel:blindProcessing	Blind Data Processing	A measure that permits specified computational processing of an Asset while preventing the human processor from directly accessing the original protected data.	dpv:TechnicalMeasure

The initial vocabulary defines three measures: Spatial Resolution Limitation, Data Minimisation and Blind Data Processing. These measures may apply to personal data, confidential information, commercially sensitive data, indigenous knowledge, ecological information and other Assets requiring controlled access or processing.

OpenREL reuses established vocabularies for technical, organisational, legal and physical measures wherever suitable concepts are available. The Data Privacy Vocabulary provides a taxonomy of technical and organisational measures, including measures for access control, security, governance and data protection. The OpenREL scheme provides additional concepts where more specific operational requirements are needed.

- OpenREL concepts MAY reference related DPV concepts using informative relationships. Such references do not necessarily establish semantic equivalence or subclass relationships.
- Where a spatial threshold is required, the applicable policy SHOULD specify the measurement model, threshold value and unit of measurement. Implementations MAY apply spatial aggregation, generalisation, coordinate precision reduction or other suitable techniques.

- Implementations MAY support selective queries, filtering of attributes or records, controlled extraction or other mechanisms that reduce unnecessary exposure of protected information.
- The applicable policy SHOULD specify the authorised data scope or the criteria used to determine the required subset.
- Data minimisation MAY be combined with other safeguards, including anonymisation, access controls, aggregation and restrictions on subsequent use.
- For ‘blind data processing’, the processing environment MAY access the original Asset, while the human processor is restricted to authorised code, execution information and permitted derived outputs.
- Implementations SHOULD apply additional controls to prevent disclosure through logs, intermediate results, derived outputs or repeated execution.
- Implementations SHOULD use the published OpenREL concept IRIs when referencing the measures defined by this scheme.
- Implementations MUST distinguish the classification of a measure from the operational conditions and parameters required to enforce it.
- Implementations MAY combine multiple measures to satisfy the requirements of a policy.

A.5.6.11 TRUST Connectors

OpenREL defines the SKOS ConceptScheme `openrel:trustType` to classify security profiles associated with trusted connectors operating in International Data Spaces (IDS).

The scheme defines three concepts corresponding to the IDS Base, Trust and Trust+ security profiles.

These concepts support policies that require a connector to satisfy specified security and trust requirements before accessing, transferring or processing an Asset, and are used to formulate constraints. The applicable LeftOperand identifies the connector security characteristic being evaluated, while the RightOperand identifies the required profile. The concepts are associated with corresponding security profile identifiers defined by the IDS Information Model.

The Base profile establishes baseline security and trust requirements. The Trust profile specifies extended security capabilities. The Trust+ profile provides enhanced security requirements, including hardware-supported security mechanisms. OpenREL references the corresponding IDS identifiers without redefining their normative requirements.

Concept Scheme IRI: openrel:trustType

Table A.5.6.11 openrel:trustType Concept Scheme

Concept IRI	Preferred label	Definition	IDS reference
openrel:base	IDS Base Security Profile	Baseline security and trust requirements for participation in the certified IDS ecosystem.	idsc:BASE_SECURITY_PROFILE
openrel:trust	IDS Trust Security Profile	Extended security requirements beyond the baseline IDS security profile.	idsc:TRUST_SECURITY_PROFILE
openrel:trust-plus	IDS Trust+ Security Profile	Enhanced trust and security requirements, including hardware-supported security mechanisms.	idsc:TRUST_PLUS_SECURITY_PROFILE

- Connector Security Profile concepts MAY be used as controlled values in ODRL Constraints or OpenREL Named Constraints.
- A policy MAY require a connector to satisfy a specified IDS security profile as a condition of accessing or processing an Asset.
- Implementations SHOULD distinguish a connector's declared security profile from verified evidence of compliance with that profile.
- Where a policy requires certification, the evaluator SHOULD establish the validity, scope and applicability of the relevant certification or assessment evidence.
- The existence of a profile declaration MUST NOT, by itself, be interpreted as proof of certification.
- Implementations SHOULD use the published OpenREL concept IRIs when referencing these classifications.
- Implementations MAY use the corresponding IDS identifiers directly where supported by the applicable policy profile and evaluation environment.

A.5.6.12 Types of Subject Rights

OpenREL defines the SKOS ConceptScheme openrel:typeSubjectRights to identify rights held by subjects in relation to data, information, knowledge and other protected resources. Subjects may include natural persons, organisations, communities and other collective rights holders. The vocabulary includes rights associated with the GDPR, as well as rights that may arise under other legal, contractual, customary or ethical frameworks.

The vocabulary contains 24 concepts. Where a GDPR provision is associated with a concept, the mapping is expressed using skos:closeMatch. Such mappings do not imply that the broader OpenREL concept is identical to the corresponding GDPR right or that the right applies universally.

Concept Scheme IRI: openrel:typeSubjectRights

Table A.5.6.12: openrel:typeSubjectRights Concept Scheme

Concept IRI	Preferred label	Definition	GDPR mapping
openrel:inform-of-collection	Right to Information about Collection	A right to be informed when information concerning a subject is collected directly from that subject, where the applicable framework requires such notice.	eu-gdpr:A13
openrel:inform-sources	Right to Information about Sources	A right to be informed about the sources of information concerning a subject when that information was obtained from other parties or sources, where required.	eu-gdpr:A14
openrel:right-access	Right to Access	A right held by a subject to access information or resources concerning that subject, subject to applicable conditions and limitations.	eu-gdpr:A15
openrel:right-rectification	Right to Rectification	A right to request correction or completion of inaccurate or incomplete information concerning a subject, where applicable.	eu-gdpr:A16
openrel:right-erasure	Right to Erasure	A right to request erasure of information concerning a subject when the applicable grounds and conditions for erasure are met.	eu-gdpr:A17

openrel:right-restriction	Right to Restriction of Processing	A right to require specified processing of information concerning a subject to be restricted under applicable conditions.	eu-gdpr:A18
openrel:right-notification	Right to Notification of Changes	A right to receive or benefit from required notifications concerning rectification, erasure, or restriction of processing of information about a subject.	eu-gdpr:A19
openrel:right-portability	Right to Data Portability	A right to receive specified information concerning a subject in a usable format and, where applicable, to transmit it to another recipient or service.	eu-gdpr:A20
openrel:right-object	Right to Object to Processing	A right to object to specified processing or use of information concerning a subject, under the applicable framework.	eu-gdpr:A21
openrel:right-non-automation	Right Not to Be Subject to Certain Automated Decisions	A right not to be subject to specified decisions based solely on automated processing, where the applicable framework provides that protection.	eu-gdpr:A22
openrel:right-intervention	Right to Human Intervention	A right to obtain human intervention in specified automated decision-making processes where required by the applicable framework.	eu-gdpr:A22-3-a
openrel:right-point-of-view	Right to Express a Point of View	A right to express a subject's position or perspective in a specified decision-making or review process.	eu-gdpr:A22-3-b
openrel:right-contest	Right to Contest a Decision	A right to challenge or contest a decision affecting a subject, where the applicable framework provides for such a challenge.	eu-gdpr:A22-3-c
openrel:right-withdraw	Right to Withdraw Consent	A right to withdraw consent previously given for specified processing or use of information or resources, where such withdrawal is permitted.	eu-gdpr:A7-3
openrel:right-complaint	Right to Lodge a Complaint	A right to lodge a complaint with an appropriate oversight or supervisory body about an alleged infringement of applicable rights.	eu-gdpr:A77
openrel:remedy-supervisory	Right to Judicial Remedy against Supervisory Decisions	A right to seek an effective judicial remedy against a decision or failure to act by a competent supervisory authority, where provided by law.	eu-gdpr:A78
openrel:remedy-infringement	Right to Judicial Remedy for Rights Infringement	A right to seek an effective judicial remedy where rights concerning a subject have allegedly been infringed, under applicable law.	eu-gdpr:A79
openrel:right-prior-consent	Right to Prior Consent	A right to require the subject's consent before specified collection, access, processing, disclosure, or reuse of protected information or resources.	-
openrel:right-withdraw-authorisation	Right to Withdraw Authorisation	A right to revoke an authorisation previously granted for specified access, use, processing, or disclosure	-

		of protected information or resources, subject to applicable limitations.	
openrel:right-benefit-sharing	Right to Benefit Sharing	A right to participate in or receive agreed benefits arising from specified use or exploitation of information, knowledge, or other resources associated with a subject.	-
openrel:right-attribution	Right to Attribution	A right to appropriate acknowledgement of a subject's authorship, contribution, custodianship, or other recognised relationship to information, knowledge, or resources.	-
openrel:right-confidentiality	Right to Confidentiality	A right to require that specified information concerning a subject be protected against unauthorised access or disclosure.	-
openrel:right-collective-decision	Right to Collective Decision-Making	A right of a collective subject to participate in, approve, or determine decisions about specified uses of resources associated with that collective subject.	-
openrel:right-restrict-disclosure	Right to Restrict Disclosure	A right to limit disclosure of specified protected information or resources to defined recipients, purposes, conditions, or contexts.	-

The concepts MAY be used to identify rights relevant to the subject of an OpenREL policy. Implementations SHOULD use the published concept IRIs and SHOULD identify the applicable framework, rights holder, scope and conditions when asserting or evaluating such rights.

Implementations MUST NOT infer from membership in this vocabulary alone that a right applies to every subject or in every jurisdiction. GDPR mappings MUST NOT be interpreted as extending GDPR rights automatically to non-person subjects.

A.5.7 Scenarios

OpenREL defines the SKOS ConceptScheme openrel:typeScenario to classify human-readable scenarios associated with the access, use, processing, reuse and dissemination of Assets.

Scenario Types provide a bridge between the language used by Asset owners, rights holders, requestors and policy administrators (e.g. end users), and the formal representation of rights and obligations through ODRL and OpenREL.

Scenario Types are organised using SKOS hierarchical relationships. The hierarchy supports scenario discovery, classification, navigation and selection in user interfaces and APIs.

Concept Scheme IRI: openrel:typeScenario

Table A.5.7 openrel:typeScenario Concept Scheme

Concept IRI	**Preferred label**	**Broader concept**	**Definition**
openrel:group-research-only	Research Only	-	Scenarios restricting use of an Asset to research or research-related activities.
openrel:academic-only	Academic Only	group-research-only	Use of an Asset is restricted to academic research activities.

openrel:confirmed-researcher	Confirmed Researcher Only	group-research-only	Use of an Asset is restricted to persons whose researcher status has been verified according to specified criteria.
openrel:confirmed-rpo	Confirmed Research Performing Organisation	group-research-only	Use of an Asset is restricted to research-performing organisations whose status has been verified according to specified criteria.
openrel:affirmed-research	Affirmed Researcher	group-research-only	A requestor declares researcher status or affiliation with a research or academic institution.
openrel:group-commercial-scenarios	Commercialisation Scenarios	-	Scenarios concerning commercial and non-commercial uses of an Asset.
openrel:commercial-use	Commercial Use Allowed	group-commercial-scenarios	Commercial use of an Asset is allowed under the applicable policy.
openrel:non-commercial-only	Non-Commercial Only	group-commercial-scenarios	Use of an Asset is restricted to non-commercial purposes, including applicable research and other activities.
openrel:group-funding-related	Funding Considerations	-	Scenarios concerning conditions arising from the funding of an Asset or associated activities.
openrel:funding-related	Funder-defined Permissions and Prohibitions	group-funding-related	Use of an Asset is subject to permissions, prohibitions or other conditions established by an applicable funding agreement or policy.
openrel:public-funding-open	Public Funding - Open Access	funding-related	An Asset produced through public funding is subject to applicable open-access or open-licensing requirements.
openrel:group-legal-ethical	Legal and Ethical Considerations	-	Scenarios concerning legal, regulatory and ethical requirements associated with the use of an Asset.
openrel:legal-considerations	Legal Considerations	group-legal-ethical	Use of an Asset is subject to applicable legislation, regulations or other legal requirements.
openrel:gdpr-compliance	GDPR Compliance	legal-considerations	Processing of personal data is subject to applicable requirements of the General Data Protection Regulation.
openrel:gdpr-controller	GDPR Controller	legal-considerations	A processing arrangement requires identification of the data controller and fulfilment of applicable

			controller responsibilities under the GDPR.
openrel:gdpr-special-category	Special Categories of Personal Data	legal-considerations	Processing of special categories of personal data is subject to the conditions and safeguards established by GDPR Article 9.
openrel:ethics-considerations	Ethics-Related Considerations	group-legal-ethical	Use of an Asset is subject to applicable ethical principles, policies, approvals or governance requirements.
openrel:consent-gdpr	GDPR Consent	legal-considerations	Processing of personal data requires valid data subject consent where consent is relied upon as the applicable lawful basis.
openrel:ethics-approval	Ethics Approval	ethics-considerations	Access to or use of an Asset requires documented approval from an authorised ethics committee or institutional review board.
openrel:explicit-consent	Explicit Consent	ethics-considerations	Specified uses or reuse activities require explicit consent from an appropriately authorised Party.
openrel:group-publication-status	Publication Status	-	Scenarios concerning publication state, embargo, withdrawal or revocation.
openrel:status-revoked	Revocation or Withdrawal	group-publication-status	Access to or use of an Asset is subject to restrictions associated with its withdrawal or revocation.
openrel:status-embargoed	Work is Embargoed	group-publication-status	Access to or use of an Asset is restricted during a defined embargo period.
openrel:group-obligations	End User Obligations and Duties	-	Scenarios concerning duties associated with accessing, using or disseminating an Asset.
openrel:attribution	Attribution	group-obligations	Use or reuse of an Asset is subject to an attribution or citation requirement.
openrel:licence-preserved	Share-Alike	group-obligations	Distribution of adapted material is subject to applicable requirements to use the same or a compatible licence.
openrel:notification-duty	Notification Duties	group-obligations	Access, use or dissemination of an Asset is subject to specified notification or reporting duties.

openrel:no-derivatives	No Derivatives	group-obligations	Dissemination of an Asset can only take place in unmodified form.
openrel:private-use	Private Modification and Use	group-obligations	An Asset can be modified for private use.
openrel:group-open-science	Open Science Considerations	-	Scenarios concerning open access, open science and repository deposit.
openrel:open-access-os	Open Access	group-open-science	An Asset is intended to be openly accessible, subject to any separately specified conditions governing reuse.
openrel:public-repository	Public Repository	group-open-science	An Asset is to be deposited or made available through a publicly accessible repository.
openrel:group-access-considerations	Access Considerations	-	Scenarios concerning conditions, limitations and safeguards governing access to Assets.
openrel:trusted-connection	Trusted Connections	group-access-considerations	Access to an Asset requires a connection satisfying specified security or trust requirements.
openrel:european-use	EU Use	group-access-considerations	Access to or use of an Asset is restricted according to a specified European Union location, residence or establishment criterion.
openrel:strict-subset	Strictly Necessary Content	group-access-considerations	Access to an Asset is limited to the subset of information necessary for a specified purpose or authorised activity.
openrel:verified-identity	Verified Identity	group-access-considerations	Access to an Asset requires verification of the requestor's identity using an accepted verification mechanism.
openrel:system-limitations	System Limitations	group-access-considerations	Access to or processing of an Asset is restricted to identified or approved systems, devices or applications.
openrel:formal-toms	Technical and Organisational Measures	group-access-considerations	Access to or processing of an Asset requires compliance with specified technical, organisational, physical or legal safeguards.
openrel:tech-limits	Technical Access Limits	group-access-considerations	Access to an Asset is subject to technical limits such as duration, number of access events or volume of data transferred.

- Scenario groups and individual scenarios MAY be associated with Named Constraints, Actions, Policies and other OpenREL resources to identify the contexts in which those resources are relevant.
- A Scenario Type MAY be associated with one or more ODRL Permissions, Prohibitions, Duties, Actions or OpenREL Named Constraints.
- Conversely, a Named Constraint or Action MAY be relevant to multiple scenarios.
- OpenREL uses the property `openrel:questionSatisfied` to associate scenarios with contextual questions, including Who, What, Why, When and How. These classifications MAY assist users in selecting relevant scenarios and identifying appropriate policy components.
- Scenario Types MAY be used to identify candidate policy components for a particular request or use case.
- Implementations MAY use scenario selections to retrieve relevant Named Constraints, Actions or policy templates.
- SKOS hierarchical relationships MUST NOT, by themselves, be interpreted as executable permission inheritance or as evidence that a policy condition has been satisfied.
- Implementations SHOULD use the published OpenREL scenario IRIs when classifying supported use cases.
- Implementations MAY associate additional policy components, examples and guidance with individual scenarios.
- Implementations MUST distinguish descriptive scenario classification from executable policy evaluation.

A.6. Left Operands

A.6.1. Scope and Normative Basis

OpenREL extends the ODRL 2.2 constraint mechanism with domain-specific LeftOperand individuals. ODRL defines odrl:LeftOperand and odrl:leftOperand; OpenREL reuses these terms and does not redefine their class/property semantics. OpenREL publishes a number of individuals, and references external terms from ODRL and IDSC that are adopted by IRI, not republished as OpenREL definitions.

The ODRL Information Model permits profiles to create additional LeftOperand instances.

A.6.2. Schema and Encoding

The ODRL Constraint requires one odrl:leftOperand (an odrl:LeftOperand individual), one odrl:operator, and an odrl:rightOperand or odrl:rightOperandReference. OpenREL adds operand individuals in its own namespace. No additional schema class or property is required solely to register these operands.

```
openrel:agentRole a odrl:LeftOperand ;
   rdfs:label "Agent Role"@en .
```

A.6.3. Registry

The companion registry publishes 20 OpenREL LeftOperand individuals and references the ODRL and IDSC inventories without copying their definitions. Some OpenREL Left operands that do not reference a concept scheme (for example, agentAffiliation that expects to resolve to an Organisation IRI) can be determined via Parameters from a specific context, this is discussed in detail in the section on Parameters.

Registry IRI: openrel:LeftOperandRegistry

Table A.6.3 Left Operand Registry (OpenREL individuals Only)

IRI	Source label	Value / scheme
accessCategory	Access Limitations	openrel:typeAccess
adaptationStatus	Adaptation Status	Adaptation status (boolean / classification)
agentAffiliation	Agent Affiliation Constraint	Organisation IRI
agentRole	Agent Role Constraint	openrel:typeAgent
assetLocation	Physical or Virtual Asset Location	Location IRI
consentRequired	Consent or Authorisation Required	openrel:typeConsent
contextLimitation	Context Limitation	openrel:typeContext
delegationToken	Delegation Token	xsd:boolean
eidasResearcherStatus	eIDAS Researcher Status	Verified researcher status
harvestedVolume	Harvested Volume	Cumulative harvested quantity
ipRange	IP Range	CIDR range
legalBasis	Legal Basis	DPV legal basis instance
locationLimitation	Location Limitation	Location or geometry
logSpecification	Logging Specification	Specification IRI
notificationDuty	Notification Duty	openrel:notification-target-type
processingInfrastructure	Processing Infrastructure	Infrastructure or execution-environment IRI
revenuePermitted	Revenue Permitted	Revenue source scheme (TRSP)
securityProfile	Security Profile	openrel:typeTrust
tomLimitation	TOM Limitation	openrel:type-tom
usageLimitation	Technical Usage Limitations	Usage limit / applicable usage classification

A.6.5 External Vocabulary Adoption

ODRL operands are adopted by their canonical odrl: IRIs. IDSC operands are adopted by their idsc: IRIs only where their operational semantics, supported operators, expected value types, and evaluation context are compatible with the intended policy.

A.6.6 Relationship to OpenREL Concept Schemes

An operand identifies a value or feature to evaluate; a SKOS concept scheme may supply permissible right-operand values. The relevant schemes include, for example, typeConsent, typeAgent, typeContext, type-tom, type-purpose, notification-target-type and the revenue source scheme in TRSP. The scheme membership of right operands is not enforced by rdfs:range alone; implementations SHOULD publish SHACL validation shapes where closed or restricted schemes are intended.

- An OpenREL LeftOperand MUST have a stable IRI, a definition of the value being evaluated, and a documented means of obtaining or calculating that value.
- An operand specification SHOULD define the applicable subject (requestor, Asset, execution environment or event), permissible operators, right-operand datatypes or vocabularies, units, and behaviour when the value cannot be established.
- The evaluator MUST distinguish absence of evidence, failed verification, and a confirmed negative value; it MUST NOT interpret an unknown value as a successful constraint.

- Where multiple conditions apply, use ODRL LogicalConstraints and their defined operators rather than encoding conjunction or disjunction in a concept label.
- The existence of a consent record, payment receipt, delegation token, or security certificate is not by itself evidence of current validity; the corresponding operand semantics MUST define how validity is evaluated.

A.6.7 References

W3C ODRL Information Model 2.2: https://www.w3.org/TR/odrl-model/
W3C ODRL Vocabulary & Expression 2.2: https://www.w3.org/TR/odrl-vocab/
W3C ODRL Profile Best Practices: https://w3c.github.io/odrl/profile-bp/

A.7 Parameters and Right Operands

A.7.1 Purpose and Scope

OpenREL Parameters provide reusable symbolic definitions for values that are not fixed when a policy template is authored. They are particularly useful when Constraints require right-operand values that depend on the Asset, requestor, platform, subject or current evaluation context.

A.7.2 Model, Classes, and Properties

A.7.2.1 Classes

openrel:Parameter is a reusable class for symbolic values resolved or supplied during policy instantiation or evaluation. Although Parameters can be used as a Right Operand, the current schema does not declare it as a subclass of odrl:RightOperand - Parameters can also represent party, asset, and contextual bindings. A Parameter definition is distinct from the value ultimately bound to it. Each Parameter MUST specify a parameterName, expectedDatatype, resolutionMethod, and multipleValues flag.

To resolve parameters, a new class, openrel:ParameterResolutionService is defined as an interface-specification resource (a subclass of schema:Service). A Parameter MAY reference it through openrel:parameterResolutionService. The interface MAY identify an OpenAPI document or another machine-actionable workflow through openrel:interfaceSpecification. The interface abstracts authentication, required inputs, execution, response extraction, and validation from policy authors. The concrete request/ response binding profile is outside the scope of this initial schema.

A service MAY resolve or compute a value, validate a supplied value, or support both. For a service that validates a parameter, a successful validation result does not necessarily supply the parameter value; evaluation must distinguish these outcomes or map the validation to an outcome.

Table A.7.2.1 OpenREL Parameter Classes

Class IRI	Superclass	Description
openrel:Parameter	-	A reusable definition of a value to be supplied or resolved during policy instantiation or evaluation.
openrel:ParameterResolutionService	schema:Service	An interface definition for a service that resolves or validates parameter values, abstracting authentication, inputs, execution and result extraction.

A.7.2.1.1 Parameter

IRI: openrel:Parameter | Superclass: N/A

A reusable definition of a value to be supplied or resolved during policy instantiation or evaluation.

A.7.2.1.1 Parameter Resolution Service

IRI: openrel:ParameterResolutionService | Superclass: schema:Service

An interface definition for a service that resolves or validates parameter values, abstracting authentication, inputs, execution and result extraction.

A.7.2.2 Properties

For a Named Constraint, openrel:applicableEndpoint provides a flexible NamedConstraint association that could also provide supplementary information about satisfying the requirements of the constraint. For Parameters, the dedicated parameterResolutionService property has the narrow domain openrel:Parameter and a precise interface range. A NamedConstraint can continue to reference an applicableEndpoint independently. Implementations may later introduce a common superclass or shared interface for both, but this is not required.

Table A.7.2.2 OpenREL Parameter-Related Properties

Property IRI	Domain → Range	Description
openrel:parameterName	openrel:Parameter → xsd:string	The symbolic name used to identify a parameter in a policy template or evaluation context.
openrel:expectedDatatype	openrel:Parameter → -	Identifies the RDF datatype IRI expected for each resolved parameter value.
openrel:multipleValues	openrel:Parameter → xsd:boolean	Indicates whether the parameter may resolve to multiple values of the expected datatype.
openrel:resolutionMethod	openrel:Parameter → skos:Concept	Identifies the binding-source concept describing the context or origin from which a parameter value is obtained.
openrel:parameterResolutionService	openrel:Parameter → openrel:ParameterResolutionService	Associates a Parameter with an optional service interface for resolving or validating its value.
openrel:operandParameter	odrl:LeftOperand → openrel:Parameter	Associates an ODRL Left Operand with an OpenREL Parameter that specifies how its contextual value is resolved during policy evaluation.
openrel:applicableEndpoint	openrel:NamedConstraint → -	Identifies an endpoint relevant to the interpretation, notification or evaluation of a Named Constraint.
openrel:interfaceSpecification	openrel:ParameterResolutionService → -	Identifies a machine-readable interface or workflow specification, such as an OpenAPI description.
openrel:servicePurpose	openrel:ParameterResolutionService → skos:Concept	Identifies whether a service resolves a value, validates a supplied value, or performs both operations.

A.7.3 Relationship to ODRL

ODRL defines odrl:rightOperand for comparison values and odrl:rightOperandReference for a reference to a value to be obtained. OpenREL implementations may resolve parameters during template instantiation, replacing a placeholder with a typed value, or at policy-evaluation time using a documented binding protocol. The mere presence of a parameter IRI does not specify how a processor retrieves its value. OpenREL previously defined openrel:applicableEndpoint that is maintained for backward compatibility, but the preferred mechanism is to define such an endpoint for a Named Constraint via the odrl:rightOperandReference predicate, indicating an openrel:Parameter. The example below shows how this is encoded:

```
# Define a named constraint with an OpenREL parameter node as a target

openrel:exampleEmbargoConstraint
   a openrel:NamedConstraint ;
   odrl:leftOperand odrl:dateTime ;
   odrl:operator odrl:gteq ;
   odrl:rightOperandReference openrel:parameterAvailablenDate .

# Define a Parameter node to describe its characteristics and resolution/ validation methods

openrel:parameterAvailableDate
   a openrel:Parameter ;
   openrel:parameterName "TARGET_AVAILABILITY_DATE" ;
   openrel:expectedDatatype xsd:dateTime ;
   openrel:resolutionMethod openrel:AssetMetadataValue ;
   openrel:parameterResolutionService <MyResolutionService> .

<MyResolutionService>
   a openrel:ParameterResolutionService ;
   openrel:interfaceSpecification <SomeWebInterface> .
```

```
One could consider, in future releases, to extend the properties of a ParameterResolutionService to
be more explicit in terms of input data (could also be a parameter, such as an Asset PID) and the
exact value to be obtained from the return of the service. Annex C defines such a binding,
resolution and evaluation specification.
```

A.7.4 Parameter Binding Sources and Context

OpenREL defines a number of contexts for the resolution of parameters. Not all of these parameters are resolved at run-time, for example, the affiliation and bona fides of a researcher is evaluated with a frequency determined by a different set of factors, and does not normally change from one request to the next. The source of a parameter binding is provided in the openrel:resolutionMethod property.

The OpenREL Parameter Binding Source concept scheme (`openrel:ParameterBindingSource`) defines a controlled vocabulary for identifying the source, context or evaluation mechanism from which a Parameter obtains its value.

The scheme is intended to support parameterised policies and Named Constraints, particularly where a RightOperand must be resolved dynamically during policy instantiation or runtime evaluation.

Concept Scheme IRI: openrel:ParameterBindingSource

Table A.7.4 openrel:ParameterBindingSource Concept Scheme

Concept	Preferred label	Definition (summary)
openrel:PolicyValue	Policy Value	A value explicitly specified in a policy or policy template.
openrel:SuppliedParameterValue	Supplied Parameter Value	A value explicitly supplied by a human or software system during the instantiation or evaluation of a parameterised policy.
openrel:RequestorContext	Requestor Context	A value obtained from the current access or usage request, including information about the requesting Party, requested action or request environment.
openrel:PartyContext	Party Context	A value obtained from established characteristics, credentials or assertions associated with a Party.
openrel:SubjectContext	Subject Context	A value obtained from the permissions, restrictions, preferences or assertions associated with a data subject or another subject whose interests are governed by a policy.
openrel:RepositoryContext	Repository Context	A value obtained from the identity, configuration, policies or administrative context of a repository.
openrel:PlatformContext	Platform Context	A value obtained from the configuration, capabilities or operational state of the platform on which an Asset is hosted, disseminated or processed.
openrel:RuntimeContext	Runtime Context	A value obtained from the current execution environment of the policy evaluator.
openrel:PolicyUsageValue	Policy Usage Value	A value obtained from the recorded history or current state of activities governed by a policy.
openrel:EventValue	Event Value	A value obtained from the occurrence, outcome or recorded state of an identifiable event.
openrel:ContractStateValue	Contract State Value	A value obtained from the current or recorded state of an agreement, contract or associated contractual workflow.
openrel:AssetMetadataValue	Asset Metadata Value	A value obtained from metadata describing the Asset or Asset Collection governed by a policy.
openrel:ExternalReferenceValue	External Reference Value	A value obtained by retrieving or querying an external information resource or service.
openrel:AutoTest	Automated Test Result	A value obtained from an automated test or verification process using explicitly defined evaluation rules.
openrel:HumanTest	Human Test Result	A value obtained from an assessment, verification or decision performed by a human evaluator.
openrel:AgentTest	AI Agent Test Result	A value obtained from an assessment or verification process performed by an AI agent.

A.7.5 Evaluation Requirements

- A processor SHOULD identify the parameter definition, retrieve its value from the declared source, validate the value against the expected datatype, and apply the constraint operator only after successful resolution.
- Access tokens and other credentials MUST NOT be published as literal values in a public policy graph. They should be resolved securely at runtime, with appropriate redaction, expiry, and access control.
- An IRI-valued parameter SHOULD use an RDF IRI where the binding contract requires a resource identifier; an xsd:anyURI literal MAY be used where the declared binding contract explicitly expects a literal.
- Implementations claiming support for OpenREL parameters MUST distinguish parameter definitions from resolved values and MUST specify the resolution semantics for every supported binding source.
- Implementations SHOULD validate expected datatypes and MUST NOT treat unresolved parameters as successful constraints.
- Failure to resolve a required parameter, datatype mismatch, expired evidence, and unauthorised source access MUST have explicit error or indeterminate outcomes.

A.7.6 Parameter Registry

OpenREL predefines a number of Parameters on the basis that these were needed for development of example and template Policy encodings, but implementers are encouraged to develop additional Parameter definitions to suit their individual needs.

Registry IRI: openrel:ParameterRegistry

Table A.7.6 OpenREL Parameter Registry

Group	IRI	Parameter Name	Label	Description
Temporal context	openrel:startDateParameter	START_DATE	Start Date	The start date for a period, usually a period of access.
Temporal context	openrel:endDateParameter	END_DATE	End Date	The end date for a period, usually a period of access.
Repository and assigner context	openrel:parameterPublisherIRI	PUBLISHER_IRI	Publisher IRI	A unique identifier for a publisher (PID, URI, URL).
Repository and assigner context	openrel:parameterAssignerIRI	ASSIGNER_IRI	Assigner IRI	A unique identifier for the assigner, such as a curator or rights owner.
Repository and assigner context	openrel:parameterTargetIRI	TARGET_IRI	Target IRI	An identifier for the target Asset or Asset Collection.

Platform context	openrel:parameterInfraIRI	INFRASTRUCTURE_IRI	Infrastructure IRI	An identifier for the infrastructure on which an Asset may be used.
Platform context	openrel:parameterVolume	VOLUME_AS_INTEGER	Maximum Download Volume	The maximum permitted quantity of data associated with executions of an Action within a defined accounting scope, such as a within a Ticket, a session, or a period.
Platform context	openrel:parameterUnit	UNIT_OF_MEASURE	Unit of Measure	A unit of measurement, potentially associated with a rate or an absolute quantity.
Platform context	openrel:parameterAuthorisedIP	AUTHORISED_IP_ARRAY	Authorised IP Addresses	A list of authorised IP addresses or address ranges.
Platform context	openrel:parameterAuthorisedIRI	AUTHORISED_IRI_ARRAY	Authorised Endpoint IRIs	A list of authorised endpoint IRIs.
Platform context	openrel:parameterClearingHouseAPI	CLEARINGHOUSE_API	Clearinghouse API	The API endpoint for clearinghouse notifications.
Platform context	openrel:parameterTokenVerificationAPI	TOKEN_VERIFICATION_API	Access Token Verification Endpoint	An API endpoint used for access token verification.
Platform context	openrel:parameterAccessToken	ACCESS_TOKEN	Access Token	An access token provided by the platform.
Requesting Party context	openrel:parameterAssigneeIRI	ASSIGNEE_IRI	Assignee IRI	An identifier for the Party receiving rights under a policy.
Requesting Party context	openrel:parameterInstitutionIRI	INSTITUTION_IRI	Institution IRI	An identifier for an institution with which the requestor is affiliated.
Requesting Party context	openrel:parameterPurposeIRI	INTENDED_PURPOSE	Intended Purpose IRI	An identifier for the intended purpose of use.
Data Subject context	openrel:parameterDataSubjectIRI	DATA_SUBJECT_IRI	Data Subject IRI	An identifier for the data subject associated with protected information.
Data Subject context	openrel:parameterInformedConsentIRI	INFORMED_CONSENT_IRI	Informed Consent Evidence IRI	An identifier for evidence documenting informed consent.
Data Subject context	openrel:parameterConsentExpiryDate	INFORMED_CONSENT_EXPIRY	Informed Consent Expiry Date	The date and time at which applicable informed consent expires.

Data Subject context	openrel:parameterDataSpecIRI	DATA_SPECIFICATION_IRI	Data Specification IRI	An identifier for a specification defining the data in scope for access.
Data Subject context	openrel:parameterPurposeVerificationIRI	PURPOSE_VERIFICATION_IRI	Purpose Verification IRI	An identifier for a manual or automated purpose-verification mechanism.
Asset metadata context	openrel:parameterPublicationDate	TARGET_PUBLICATION_DATE	Asset Publication Date	The publication date of an Asset, potentially used for embargo calculations.
Asset metadata context	openrel:parameterLicenceList	TARGET_LICENCE_LIST	Asset Licence List	Identifiers for the licences applicable to an Asset or Asset Collection.
Temporal context	openrel:parameterCalculatedStartDate	CALCULATED_START_DATE	Calculated Start Date	A start date calculated at runtime from one or more input Parameters using a designated Parameter Resolution Service.
Requesting Party context	openrel:parameterConsentAuthorisation	CONSENT_AUTHORISATION	Consent or Authorisation Verification	The verified consent or authorisation type applicable to the current request, resolved through an authorisation verification service.
Platform context	openrel:parameterUsageCount	MAX_USAGE_COUNT	Maximum Usage Count	The maximum permitted number of executions of an Action within a defined accounting scope and counting period.
Temporal context	openrel:parameterEmbargoDuration	EMBARGO_DURATION	Embargo Duration	The duration of an embargo period, supplied by the repository or platform for a specified Asset or Asset Collection.
Requesting Party context	openrel:parameterEidasResearcherStatus	EIDAS_RESEARCHER_STATUS	eIDAS Researcher Status	A Boolean indicating whether the requestor has an eIDAS-verified identity and an accepted researcher-status assertion associated with that identity.
Asset metadata context	openrel:parameterAdaptationStatus	ADAPTATION_STATUS	Adaptation Status	A Boolean indicating whether the asset being evaluated constitutes an adaptation of the original licensed material under the applicable licence.

Annex B: Policy Composition from Pairwise Comparison

OpenREL CSupplementary Material - version 0.5
6 November 2026.

Supplementary specification section for Annex A, Section 4 - Comparison and Evaluation. The authoritative schema and vocabulary are not modified by this document.

B.1. Scope and interpretation

This section specifies how evidence from pairwise comparisons of ODRL Actions, Permissions, Prohibitions, and Duties can contribute to a combined policy and to an aggregate evaluation outcome. The combination is context-specific: assets, parties, requested actions, constraints, and the fulfilment of duties affect the result. A PolicyCollection identifies the input policies; it is not automatically the resulting combined policy.

In the examples, we refer to input policies defined in a PolicyCollection as ‘A’ and ‘B’. The policy combination ‘C’ is the conjunction of the applicable and compatible requirements of A and B. It need not be identical to either source policy. Restrictions may originate from both sources. “Most restrictive” means the effective intersection of permitted behaviours, not selecting one dominant rule from each source.

B.2. Existing OpenREL Resources

The specification makes use of the following OpenREL schema resources:

Table B.2 OpenREL Resources Used for Evaluation

Resource	Existing meaning / use
openrel:PolicyCollection	Groups two or more input odrl:Policy resources via openrel:hasPolicy.
openrel:EvaluationDetail	Directed comparison record; not itself an aggregate decision.
openrel:EvaluationMapping	Reusable subtype of EvaluationDetail containing comparison evidence.
openrel:evaluationOutcome	Pairwise semantic relationship, e.g. compatible, contradicts, allows, includedIn, implies or missing.
openrel:policyEvaluationOutcome	Aggregate decision on PolicyCollection; policyPermit, policyConditional or policyDeny.
openrel:hasEvaluationDetail / openrel:hasEvaluation	Associate evaluation evidence with PolicyCollection.

B.3. Role-sensitive Combination Rules

OpenREL defines a number of rules for the combination of two or more policies. These combinations are often directional (if Policy A allows all of the rules in Policy B, the converse may not be true) and are dependent on the role of an Action (Permission, Prohibition, Duty and its subclasses) and any relevant constraints. These rules form the foundation of the policy combination and comparison processes defined in the Annex.

Table B.3 OpenREL Policy-Level Rules for Combining Two or More Policies

Role	Combination Principle	Evaluation Caveat
Permission	Keep actions authorised by both policies for the intended activity; intersect authorised scopes.	Two different permissions are not contradictory merely because their Actions differ.

Prohibition	Enforce every applicable prohibition from either policy.	A prohibited optional activity can be removed from C without denying the whole combination.
PreDuty	Conjoin mandatory preconditions and require fulfilment before the relevant action.	Unfulfilled but satisfiable prerequisites support CONDITIONAL; impossible ones cause DENY.
PostDuty	Retain all applicable obligations that follow an allowable action.	A feasible post-duty can accompany PERMIT; its future fulfilment must remain traceable.
Constraint	Intersect admissible values or scopes and preserve incomparable restrictions jointly.	An empty intersection for a required activity causes DENY.

B.4. Pairwise Relationship Interpretation

The semantic relationship between resources is evidence, not an outcome by itself. The following table gives candidate effects, assuming applicable scopes. The source and target of includedIn and implies MUST remain distinguishable, since these are directional. Table B.4 is a partial extract of the full range of outcome concepts listed in Table A.4.5.1.2.

Table B.4 OpenREL Action and Constraint Pairwise Comparison Outcomes

Pairwise relationship	**Candidate effect on C**	**When policy DENY follows**
openrel:compatible	Retain jointly satisfiable requirements.	Not from this relationship alone.
openrel:includedIn openrel:implies	Use the stated direction to determine whether one requirement entails another; remove redundant restrictions only when justified.	When the required direction is not satisfiable and no alternative remains.
openrel:allows	Use as evidence for the stated rule-role combination; check its specific OpenREL definition.	Only if other applicable evidence establishes a mandatory conflict.
openrel:contradicts	Exclude the conflicting optional activity or identify an unsatisfiable mandatory requirement.	When contradiction affects a required action or non-waivable duty.
openrel:missing	Mark the expected rule/evidence as absent; determine whether it is mandatory.	When absence prevents satisfaction of a mandatory requirement.

B.5. Decision Procedure and Precedence

1. Determine evaluation context and mandatory requested activities. Identify all applicable input policies, rule scopes, and relevant evidence.
2. Compare applicable rules and constraints. Record directed relationships using EvaluationDetail / EvaluationMapping, with provenance and rationale, or reuse existing EvaluationMappings.
3. Construct candidate C: intersect effective permissions, apply all prohibitions, conjoin duties and constraints, and preserve non-comparable restrictions.
4. Evaluate satisfiability of every mandatory activity and obligation. If any definitive applicable conflict makes a mandatory requirement unsatisfiable, assign policyDeny.
5. Otherwise, if mandatory preconditions remain outstanding but can be satisfied, assign policyConditional.
6. Otherwise, if all mandatory preconditions are satisfied and the required activity is authorised, assign policyPermit. Record outstanding PostDuties separately.
7. If mandatory evidence is missing or evaluation coverage is incomplete, do not infer policyPermit. Treat as unresolved pending a defined completeness policy; do not silently equate missing evidence with denial.

A pairwise DENY propagates to policy-level DENY only if it is definitive, applicable, and blocks a mandatory requirement. For a single explicitly requested action, a conflict that blocks that action therefore causes DENY. For composition of a broader policy, optional conflicting actions can be omitted from C.

B.6. Illustrative Outcome Matrix

Table B.6 provides examples of the typical comparison outcomes when comparing policies. Policy A and Policy B are input policies, and Policy C is an intersection of what the minimum remaining rules should be.

Table B.6 Matrix of Illustrative Comparison Outcomes

Policy A	Policy B	Effective C	Aggregate outcome
Permits read and distribute	Permits read; prohibits distribute	Read only	PERMIT for read; DENY for a request to distribute
Requires authentication	Requires acceptance of licence	Both prerequisites	CONDITIONAL until both are satisfied
Retention ≤ 90 days	Retention ≤ 30 days	Retention ≤ 30 days	PERMIT if all other requirements hold
Requires public distribution	Prohibits public distribution	No valid required distribution	DENY
Requires attribution after use	Requires usage reporting after use	Both PostDuties	PERMIT if other requirements hold; both duties retained

B.7. Worked RDF Example

The following records a collection and its aggregate outcome using existing schema terms. The composed-policy link is an optional proposed addition defined in the accompanying Turtle patch. The example does not claim that a reasoner can derive the decision without an evaluation process.

```
@prefix openrel: <http://www.w3id.org/ns/openrel/0/> .
@prefix odrl: <http://www.w3.org/ns/odrl/2/> .
@prefix ex: <https://example.org/policies/> .

ex:comparison-1 a openrel:PolicyCollection ;
    openrel:hasPolicy ex:policy-A, ex:policy-B ;
    openrel:policyEvaluationOutcome openrel:policyPermit ;
    openrel:resultingPolicy ex:policy-C .

ex:policy-C a odrl:Policy .
# The complete effective rules of ex:policy-C must be encoded separately.
```

B.8. Normative Requirements

- PC-01. An implementation MUST distinguish pairwise semantic relationships from policy-level decisions.
- PC-02. A composition process MUST evaluate applicable Permissions, Prohibitions, mandatory Duties, and Constraints within the specified evaluation context.
- PC-03. An implementation MUST preserve directionality for includedIn, implies, and other asymmetric mappings.

- PC-04. A policy-level DENY MUST be returned when a definitive applicable conflict makes a mandatory requested activity or mandatory obligation unsatisfiable.
- PC-05. An implementation MUST NOT propagate an optional-activity conflict to a policy-level DENY solely because a pairwise conflict exists.
- PC-06. A combined policy MUST retain all applicable non-redundant restrictions and obligations from its contributing policies.
- PC-07. A policy-level PERMIT MUST NOT be inferred solely from the absence of recorded conflicts when mandatory comparison evidence is incomplete.
- PC-08. A policy-level CONDITIONAL SHOULD identify outstanding satisfiable preconditions and PreDuties.
- PC-09. A derived policy MUST retain provenance linking it to its contributing policies and evaluation evidence.
- PC-10. An implementation MAY record the derived policy with openrel:resultingPolicy.

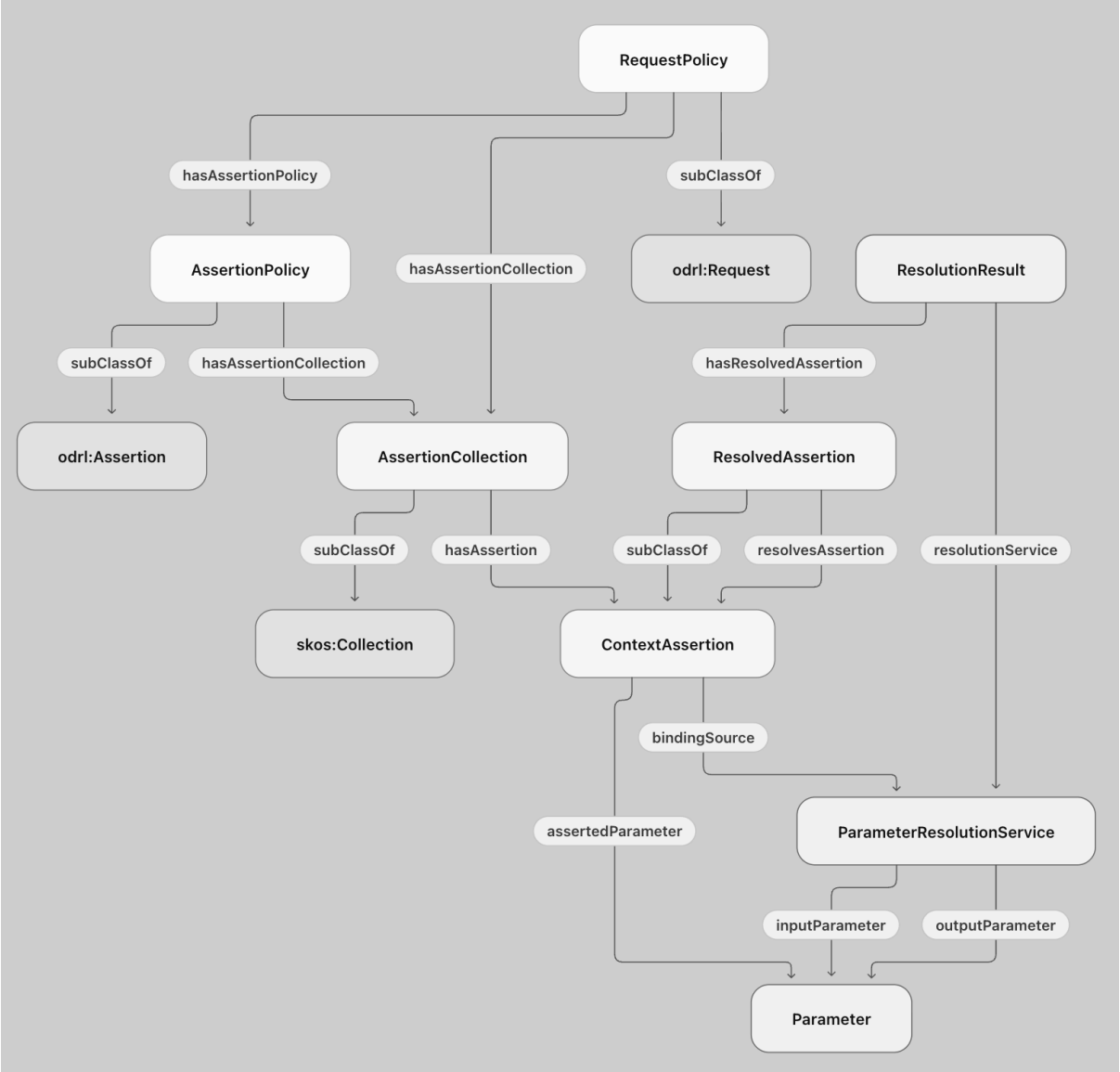
Annex C. Contextual Assertions and Parameter Resolution

OpenREL Context Extension - draft specification version 0.1
6 November 2026.

C.1. Context and Scope

This annex proposes a specification for the binding, resolution, and verification of Parameters at the point where potentially multiple Assertions (Contexts) are evaluated and compared to one or more Policies to determine compatibility.

Figure C.1 Parameter Resolution Information Model



C.2. Classes and Properties

OpenREL distinguishes reusable statements of context from requests that rely on those statements. The context types applicable to parameter binding and resolution are defined in Table A.7.4 - the openrel:ParameterBindingSource concept scheme, and is not repeated here.

An AssertionPolicy is an ODRL Assertion containing one or more collections of contextual claims. A RequestPolicy is an ODRL Request that references at least one reusable AssertionPolicy and may add request-specific assertions. This supports reuse of established evidence without requiring each request to recreate the same assertions.

A ContextAssertion binds a previously defined openrel:Parameter to an RDF value, or identifies a ParameterResolutionService that can obtain or validate the value. Resolution produces distinct ResolvedAssertion resources, grouped in a ResolutionResult. These records support provenance and later policy evaluation; they do not themselves grant access.

The extension is additive and depends on the core OpenREL Parameter and ParameterResolutionService definitions. Policy comparison and the final PERMIT / CONDITIONAL / DENY decision are outside this section.

> Annex C handles the case where variable constraints and parameters defined in one or more policies are resolved and verified. Annex B is intended as recommended procedures for processing the comparisons and evaluations required in specific cases after the values have been resolved.

C.2.1 Class Summary

Table C.2.1 OpenREL Parameter Resolution and Contextual Assertion Classes

Class	Superclass	Definition
openrel:AssertionPolicy	odrl:Assertion	Reusable OpenREL Assertion Policy with references to contextual assertions.
openrel:RequestPolicy	odrl:Request	ODRL Request augmented with reusable assertion policies and request-specific context.
openrel:AssertionCollection	skos:Collection	A collection of context assertions associated with a request or assertion policy.
openrel:ContextAssertion	rdfs:Resource	A contextual claim binding a Parameter to a value or a resolution source.
openrel:ResolvedAssertion	openrel:ContextAssertion	A contextual assertion produced by a parameter resolution operation.
openrel:ResolutionResult	rdfs:Resource	An auditable record of one parameter resolution service invocation.

C.2.2 Property summary

Property	Domain	Range	Definition
openrel:hasAssertionPolicy	openrel:RequestPolicy	openrel:AssertionPolicy	Associates a request with a reusable assertion policy.
openrel:hasAssertionCollection	owl:unionOf(AssertionPolicy, RequestPolicy)	openrel:AssertionCollection	Associates an ODRL policy with contextual assertions.
openrel:hasAssertion	openrel:AssertionCollection	openrel:ContextAssertion	Identifies a contextual assertion in a collection.
openrel:assertedParameter	openrel:ContextAssertion	openrel:Parameter	Identifies the Parameter described by a contextual assertion.

openrel:assertedValue	openrel:ContextAssertion	rdfs:Resource	Provides a directly asserted or resolved RDF value.
openrel:bindingSource	openrel:ContextAssertion	openrel:ParameterResolutionService	Identifies the service to resolve or validate an assertion.
openrel:resolvesAssertion	openrel:ResolvedAssertion	openrel:ContextAssertion	Links a resolved assertion to the assertion it resolves.
openrel:hasResolvedAssertion	openrel:ResolutionResult	openrel:ResolvedAssertion	Identifies an output assertion of a resolution operation.
openrel:resolutionService	openrel:ResolutionResult	openrel:ParameterResolutionService	Identifies the service invoked for a resolution result.
openrel:inputParameter	openrel:ParameterResolutionService	openrel:Parameter	Identifies a Parameter required as service input.
openrel:outputParameter	openrel:ParameterResolutionService	openrel:Parameter	Identifies a Parameter potentially produced by a service.
openrel:expectedClass	openrel:Parameter	rdfs:Class	Identifies the expected RDF class of an IRI-valued Parameter.

C.2.3 Class descriptions

C.2.3.1 Assertion Policy

IRI: `openrel:AssertionPolicy` | Superclass: `odrl:Assertion`

A reusable ODRL Assertion Policy that records contextual claims through references to collections of ContextAssertions. An AssertionPolicy MAY be associated with multiple requests, allowing contextual information to be reused without duplication. An AssertionPolicy records claims about the relevant context but does not, by itself, grant access to an Asset or authorise execution of an Action.

C.2.3.2 Request Policy

IRI: `openrel:RequestPolicy` | Superclass: `odrl:Request`

An ODRL Request that incorporates references to reusable AssertionPolicies and request-specific contextual assertions. A RequestPolicy provides contextual information relevant to the evaluation of a requested Action. The policy evaluator determines which rules are applicable and which parameters or assertions require resolution before an evaluation outcome can be established.

C.2.3.3 Assertion Collection

IRI: `openrel:AssertionCollection` | Superclass: `skos:Collection`

A collection of ContextAssertions associated with an AssertionPolicy or RequestPolicy. Membership is expressed through the openrel:hasAssertion property, and individual ContextAssertions MAY be shared between multiple collections. An AssertionCollection provides a grouping mechanism for contextual claims and is not, by itself, an ODRL Policy.

C.2.3.4 Context Assertion

IRI: `openrel:ContextAssertion` | Superclass: `rdfs:Resource`

A contextual claim that associates an OpenREL Parameter with an asserted value or a source from which the value can be obtained or validated. A ContextAssertion identifies the relevant Parameter through openrel:assertedParameter. A directly specified value is expressed through openrel:assertedValue, while an assertion requiring resolution MAY identify a ParameterResolutionService through openrel:bindingSource. The associated Parameter determines the expected datatype or RDF class and the permitted multiplicity of values.

C.2.3.5 Resolved Assertion

IRI: `openrel:ResolvedAssertion` | Superclass: `openrel:ContextAssertion`

A ContextAssertion produced by a parameter resolution operation, containing a value obtained or validated through the resolution process. A ResolvedAssertion references the original ContextAssertion through openrel:resolvesAssertion and provides its resolved value through openrel:assertedValue. ResolvedAssertions SHOULD be retained separately from their original assertions to preserve the distinction between the original claim and the result of resolution, and to support provenance and historical records.

C.2.3.6 Resolution Result

IRI: `openrel:ResolutionResult` | Superclass: `rdfs:Resource`

An auditable record of an invocation of a ParameterResolutionService to resolve or validate one or more contextual assertions. A ResolutionResult identifies the service invoked through openrel:resolutionService and MAY reference multiple ResolvedAssertions through openrel:hasResolvedAssertion. A resolution operation MAY produce partial results where only some of the expected parameter values can be determined. Additional provenance information MAY be recorded using PROV-O to support traceability and subsequent evaluation.

C.2.4 Parameter Binding Context

The OpenREL ParameterBindingSource vocabulary identifies the sources, contexts, and evaluation mechanisms from which parameter values are obtained during policy instantiation or runtime evaluation.

A parameter may have a value explicitly defined in a policy, supplied by a requestor, retrieved from an established information source, or determined through an evaluation process. The distinction is important because different sources have different requirements for resolution, validation, provenance, and trust.

The vocabulary is used through the openrel:resolutionMethod property of openrel:Parameter. It identifies how a value is obtained, not the value itself.

2.4.1 Parameter Binding Source Vocabulary

Concept scheme: openrel:ParameterBindingSource · Version 0.2 · 16 concepts

Concept IRI	Preferred label	Definition
`openrel:PolicyValue`	Policy Value	A value explicitly specified in a policy or policy template.

openrel:SuppliedParameterValue	Supplied Parameter Value	A value explicitly supplied by a human or software system during the instantiation or evaluation of a parameterised policy.
openrel:RequestorContext	Requestor Context	A value obtained from the current access or usage request, including information about the requesting Party, requested action or request environment.
openrel:PartyContext	Party Context	A value obtained from established characteristics, credentials or assertions associated with a Party.
openrel:SubjectContext	Subject Context	A value obtained from the permissions, restrictions, preferences or assertions associated with a data subject or another subject whose interests are governed by a policy.
openrel:RepositoryContext	Repository Context	A value obtained from the identity, configuration, policies or administrative context of a repository.
openrel:PlatformContext	Platform Context	A value obtained from the configuration, capabilities or operational state of the platform on which an Asset is hosted, disseminated or processed.
openrel:RuntimeContext	Runtime Context	A value obtained from the current execution environment of the policy evaluator.
openrel:PolicyUsageValue	Policy Usage Value	A value obtained from the recorded history or current state of activities governed by a policy.
openrel:EventValue	Event Value	A value obtained from the occurrence, outcome or recorded state of an identifiable event.
openrel:ContractStateValue	Contract State Value	A value obtained from the current or recorded state of an agreement, contract or associated contractual workflow.
openrel:AssetMetadataValue	Asset Metadata Value	A value obtained from metadata describing the Asset or Asset Collection governed by a policy.
openrel:ExternalReferenceValue	External Reference Value	A value obtained by retrieving or querying an external information resource or service.
openrel:AutoTest	Automated Test Result	A value obtained from an automated test or verification process using explicitly defined evaluation rules.
openrel:HumanTest	Human Test Result	A value obtained from an assessment, verification or decision performed by a human evaluator.
openrel:AgentTest	AI Agent Test Result	A value obtained from an assessment or verification process performed by an AI agent.

2.4.2 Relationship to Contextual Assertions and Resolution

The vocabulary distinguishes several mechanisms relevant to the OpenREL contextual assertion model.

Direct values (PolicyValue, SuppliedParameterValue) can be bound without invoking a predefined external resolution source, although supplied values may still require validation.

Contextual values (RequestorContext, PartyContext, SubjectContext, RepositoryContext, PlatformContext, RuntimeContext) are obtained from the relevant operational or administrative context. Some may be represented by reusable AssertionPolicies, while others are specific to an individual RequestPolicy.

Recorded and referenced values (PolicyUsageValue, EventValue, ContractStateValue, AssetMetadataValue, ExternalReferenceValue) may require retrieval from metadata, event records, registries, logs, or external services.

Evaluated values (AutoTest, HumanTest, AgentTest) are produced by verification or assessment processes. Their results should be distinguished from directly asserted facts and may require supporting evidence, confidence information, or human review.

A ParameterResolutionService may resolve one or more parameters using these mechanisms. The resulting values can be recorded as ResolvedAssertion instances, preserving their relationship to the original contextual assertions.

2.4.3 Example: Identifying the Binding Context

```
@prefix openrel: <http://www.w3id.org/ns/openrel/0/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

openrel:parameterInstitutionIRI
    a openrel:Parameter ;
    openrel:parameterName "institutionIRI" ;
    openrel:expectedDatatype xsd:anyURI ;
    openrel:multipleValues false ;
    openrel:resolutionMethod openrel:PartyContext .
```

This example identifies the expected parameter and its binding context. It does not itself supply an institutional identifier or establish that the requesting Party's affiliation has been verified.

2.4.4 Normative Statements

These are proposed specification requirements derived from the vocabulary's intended use, rather than requirements explicitly encoded by the vocabulary itself.

- A parameter using openrel:resolutionMethod MUST reference a recognised binding-source concept.
- A resolution process MUST distinguish the source or mechanism used to obtain a value from the value returned.
- A resolved value MUST conform to the datatype and multiplicity constraints declared for its parameter.
- A resolution process SHOULD preserve evidence of the source, time, and method used to obtain or validate a value.
- Results produced by automated, human, or AI-agent assessments SHOULD preserve the relevant evaluation evidence and provenance.
- An AI-agent assessment MUST NOT be treated as an independently verified fact solely because it was produced by a resolution service.

C.3. SHACL Validation

SHACL shapes enforce structural requirements for policy references, collections, contextual assertions, resolved outputs, and resolution services. The following table summarises the stated requirements; the SHACL file remains authoritative for their executable form.

Target class	Property / test	Cardinality	Meaning
AssertionPolicy	hasAssertionCollection	1..*	Each reusable assertion policy has a collection.
RequestPolicy	hasAssertionPolicy	1..*	At least one reusable assertion policy is required.
RequestPolicy	hasAssertionCollection	0..*	Request-specific collections are optional.
AssertionCollection	hasAssertion	1..*	Each collection contains at least one contextual assertion.
ContextAssertion	assertedParameter	1	Exactly one Parameter per assertion.
ContextAssertion	assertedValue OR bindingSource	1..*	A value or a resolution service must be supplied.
ContextAssertion	bindingSource	0..1	At most one binding service per assertion.
ResolvedAssertion	resolvesAssertion	1	Links output to the source assertion.
ResolvedAssertion	assertedValue	1..*	A resolved assertion must carry a value.
ResolutionResult	resolutionService	1	Records the service invoked.
ResolutionResult	hasResolvedAssertion	0..*	Partial or empty outputs are permitted.
ParameterResolutionService	outputParameter	1..*	A service declares one or more potential outputs.
Parameter	expectedClass	0..1	An optional expected RDF class for IRI-valued parameters.

Three SPARQL-based checks additionally address single-valued parameters, literal datatypes, and the rdf:type of IRI-valued values. These are validation constraints, not automatic instructions to invoke remote services.

```
openrel:RequestPolicyShape a sh:NodeShape ;
  sh:targetClass openrel:RequestPolicy ;
  sh:property [
    sh:path openrel:hasAssertionPolicy ;
    sh:class openrel:AssertionPolicy ;
    sh:minCount 1
  ] ;
  sh:property [
    sh:path openrel:hasAssertionCollection ;
    sh:class openrel:AssertionCollection
  ] .
```

C.4. Information Model and Processing

A RequestPolicy references one or more AssertionPolicies. Both policy types may point to AssertionCollections; each collection contains ContextAssertions. A ContextAssertion identifies a Parameter and either a direct value or a bindingSource. A ParameterResolutionService declares its input and output Parameters. ResolutionResult records the invoked service and links to one or more ResolvedAssertions, each of which identifies the ContextAssertion it resolves.

Processing sequence: (1) collect reusable and request-specific assertions; (2) identify values already supplied; (3) select the authorised service for unresolved assertions; (4) obtain service outputs; (5) validate values against Parameter expectations and the SHACL shapes; (6) preserve resolution

records for downstream evaluation. Multiple inputs to and multiple outputs from authorised services are supported by the schema.

C.5. Predefined individuals

No predefined individuals are declared yet with the exception of a ResolutionMethod. Concrete AssertionPolicy, RequestPolicy, ContextAssertion, or ResolutionResult individuals must still be developed.

C.6. Related Vocabularies

The context extension reuses the existing OpenREL Parameter definitions and the core binding-source vocabulary through openrel:resolutionMethod. The new openrel:expectedClass property identifies an RDF class rather than a vocabulary concept scheme.

C.7. Worked Turtle Examples

C.7.1 Reusable Verified Institutional Affiliation

The following example assumes the referenced Parameter and service are declared elsewhere. The IRI of the institution is used as a resource value.

```
@prefix openrel: <http://www.w3id.org/ns/openrel/0/> .
@prefix ex: <https://example.org/openrel/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

ex:affiliationPolicy a openrel:AssertionPolicy ;
  openrel:hasAssertionCollection ex:affiliationContext .

ex:affiliationContext a openrel:AssertionCollection ;
  openrel:hasAssertion ex:affiliationClaim .

ex:affiliationClaim a openrel:ContextAssertion ;
  openrel:assertedParameter ex:institutionParameter ;
  openrel:assertedValue <https://ror.org/04wxnsj81> .

ex:institutionParameter a openrel:Parameter ;
  openrel:expectedClass ex:ResearchOrganisation .
```

For the expectedClass test to pass, the validation graph must contain suitable rdf:type evidence for the referenced organisation, or an appropriate subclass relationship. The illustrative ROR IRI is not sufficient evidence on its own.

C.7.2 Request-Specific Context - Unresolved Claim

This example uses a service to verify researcher credentials via eIDAS.

```
ex:request1 a openrel:RequestPolicy ;
  openrel:hasAssertionPolicy ex:affiliationPolicy ;
  openrel:hasAssertionCollection ex:requestContext .

ex:requestContext a openrel:AssertionCollection ;
  openrel:hasAssertion ex:statusClaim .

ex:statusClaim a openrel:ContextAssertion ;
```

```
  openrel:assertedParameter ex:researcherStatusParameter ;
  openrel:bindingSource ex:eidasStatusResolver .

ex:eidasStatusResolver a openrel:ParameterResolutionService ;
  openrel:inputParameter openrel:parameterInstitutionIRI ;
  openrel:inputParameter openrel:parameterAssigneeIRI ;
  openrel:outputParameter openrel:parameterEidasResearcherStatus .
```

C.7.3 A Resolution Invocation Producing a Separate Output Assertion

```
ex:result1 a openrel:ResolutionResult ;
  openrel:resolutionService ex:statusResolver ;
  openrel:hasResolvedAssertion ex:resolvedStatus .

ex:resolvedStatus a openrel:ResolvedAssertion ;
  openrel:resolvesAssertion ex:statusClaim ;
  openrel:assertedParameter ex:researcherStatusParameter ;
  openrel:assertedValue true .
```

A ResolvedAssertion is a subclass of ContextAssertion; it must therefore also satisfy the ContextAssertion shape. Its original unresolved claim is retained, rather than overwritten.

C.8. Conformance Requirements

- CTX-001. An openrel:RequestPolicy MUST reference at least one openrel:AssertionPolicy using openrel:hasAssertionPolicy.
- CTX-002. An openrel:AssertionPolicy MUST reference at least one openrel:AssertionCollection.
- CTX-003. Each openrel:AssertionCollection MUST contain at least one openrel:ContextAssertion.
- CTX-004. Each openrel:ContextAssertion MUST identify exactly one openrel:Parameter and MUST supply an assertedValue or bindingSource.
- CTX-005. A ContextAssertion MUST NOT identify more than one bindingSource under the supplied SHACL shape.
- CTX-006. A ResolvedAssertion MUST identify the ContextAssertion it resolves and MUST supply at least one assertedValue.
- CTX-007. A ResolutionResult MUST identify exactly one resolutionService; it MAY record multiple resolved assertions.
- CTX-008. A ParameterResolutionService MUST declare at least one outputParameter; it MAY declare multiple inputParameters and outputParameters.
- CTX-009. Where expectedClass is declared, validation MUST check compatible rdf:type evidence in the validation graph, rather than assuming that an IRI alone establishes class membership.
- CTX-010. Resolution services SHOULD be registered or otherwise authorised, and implementations SHOULD retain provenance of resolution results.
- CTX-011. A ContextAssertion or AssertionPolicy MUST NOT be interpreted as an access authorisation by itself.